

BFS Traversal

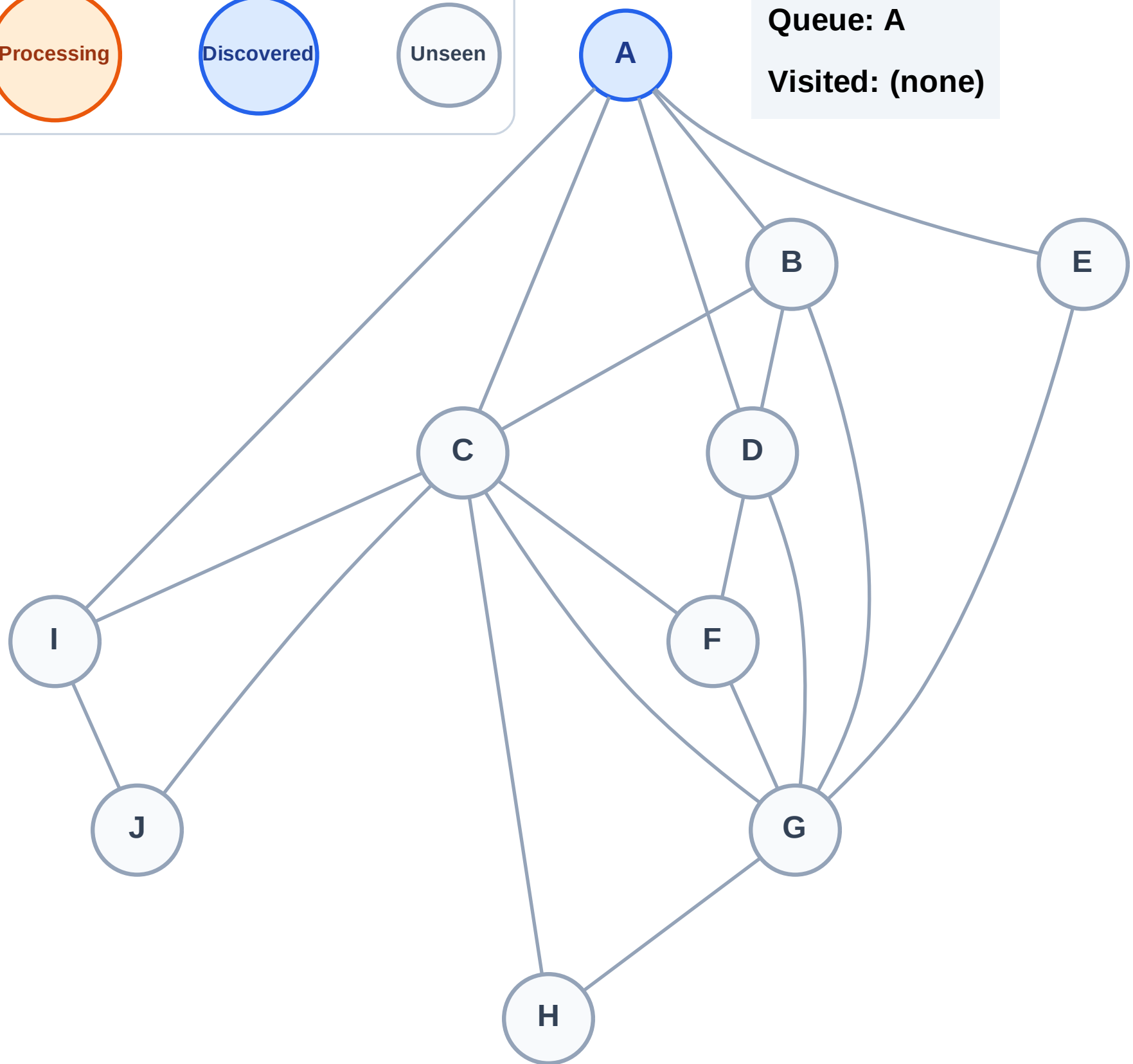
Step 1: Start at A

Legend



Queue: A

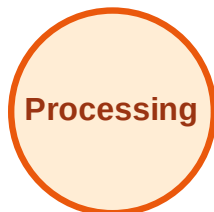
Visited: (none)



BFS Traversal

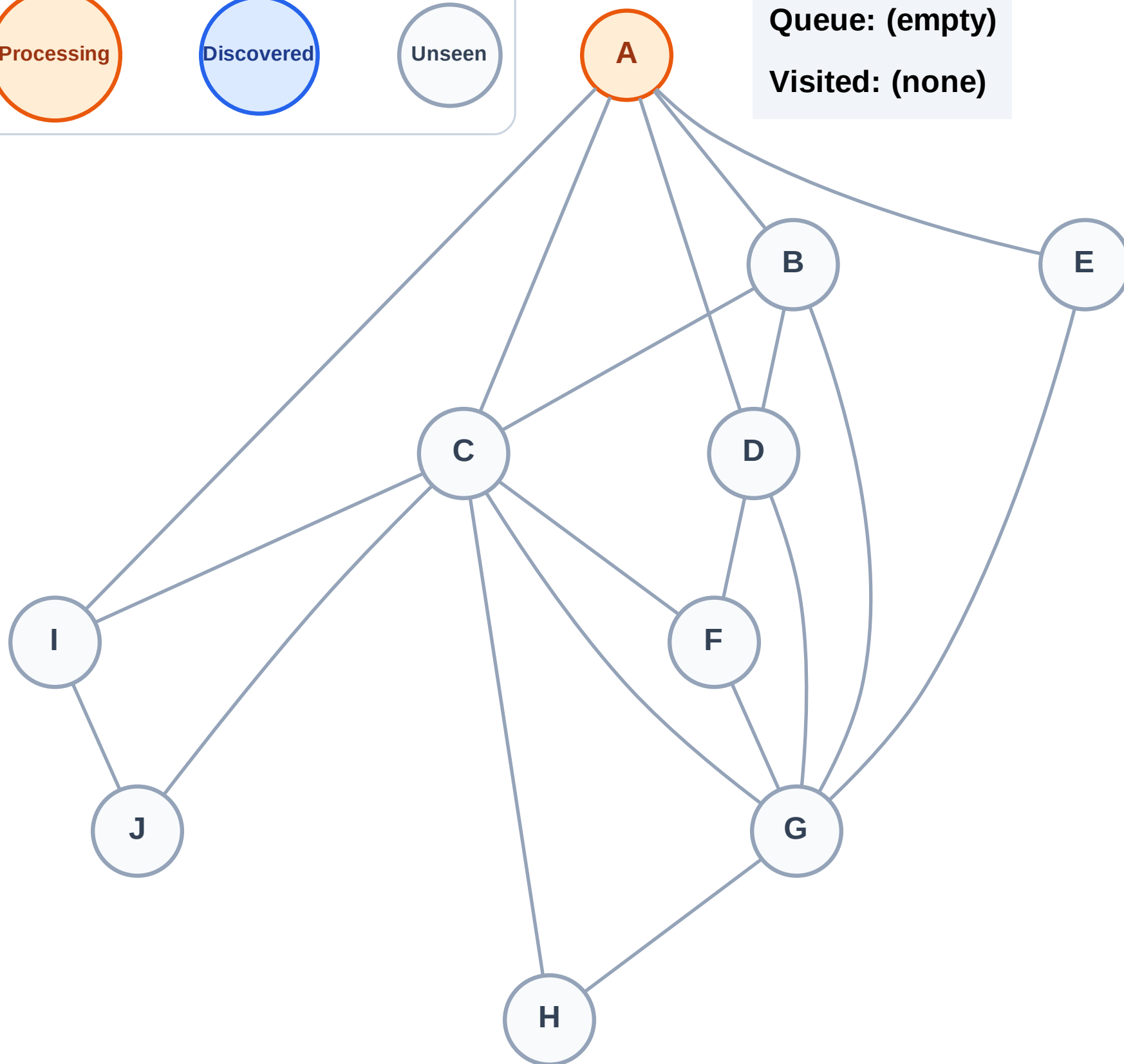
Step 2: Process node A

Legend



Queue: (empty)

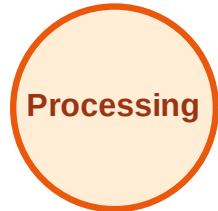
Visited: (none)



BFS Traversal

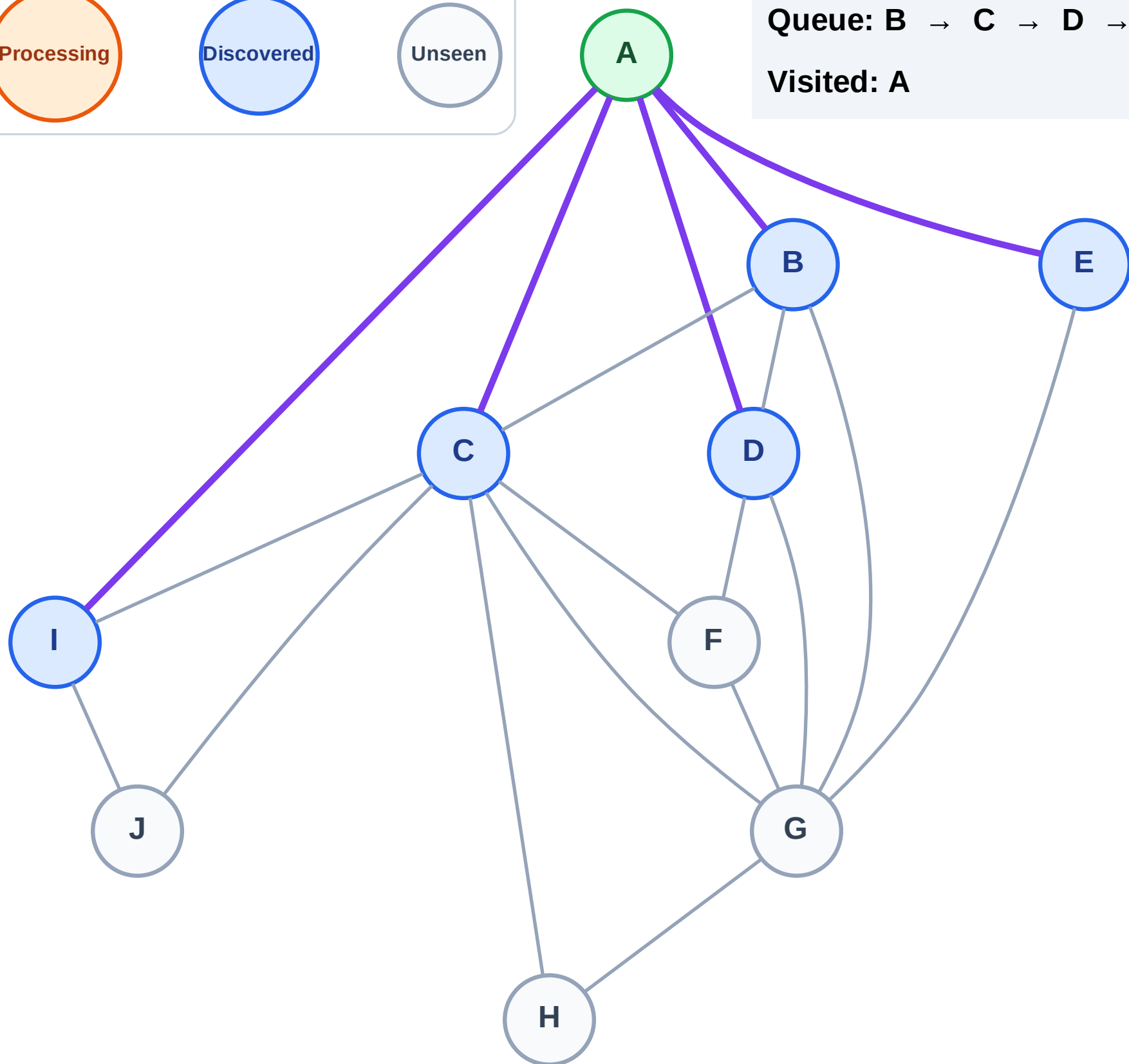
Step 3: Discover B, C, D, E, I from A

Legend



Queue: B → C → D → E → I

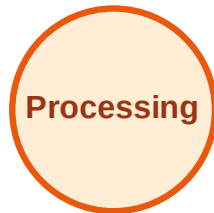
Visited: A



BFS Traversal

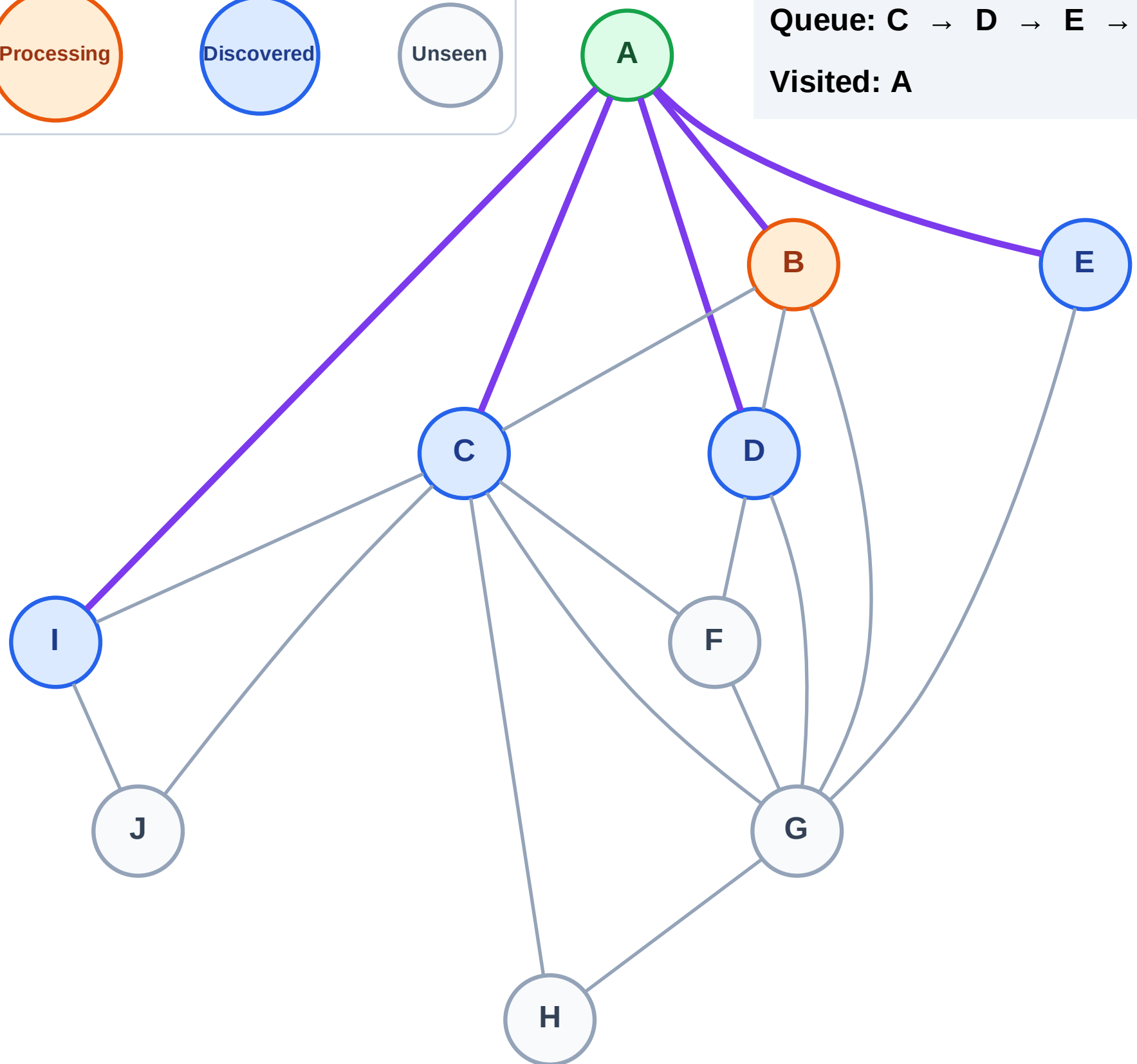
Step 4: Process node B

Legend



Queue: C → D → E → I

Visited: A



BFS Traversal

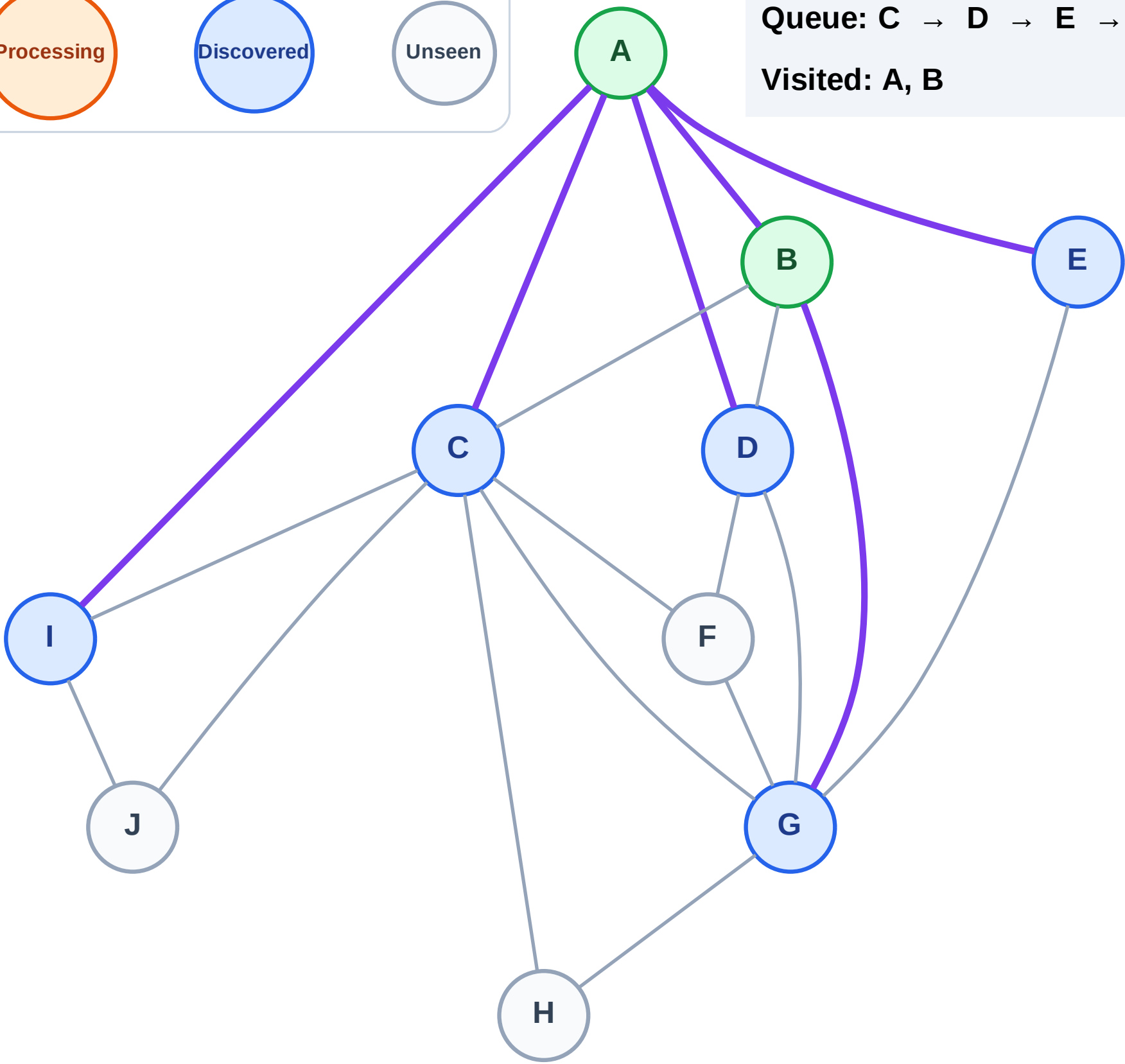
Step 5: Discover G from B

Legend



Queue: C → D → E → I → G

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

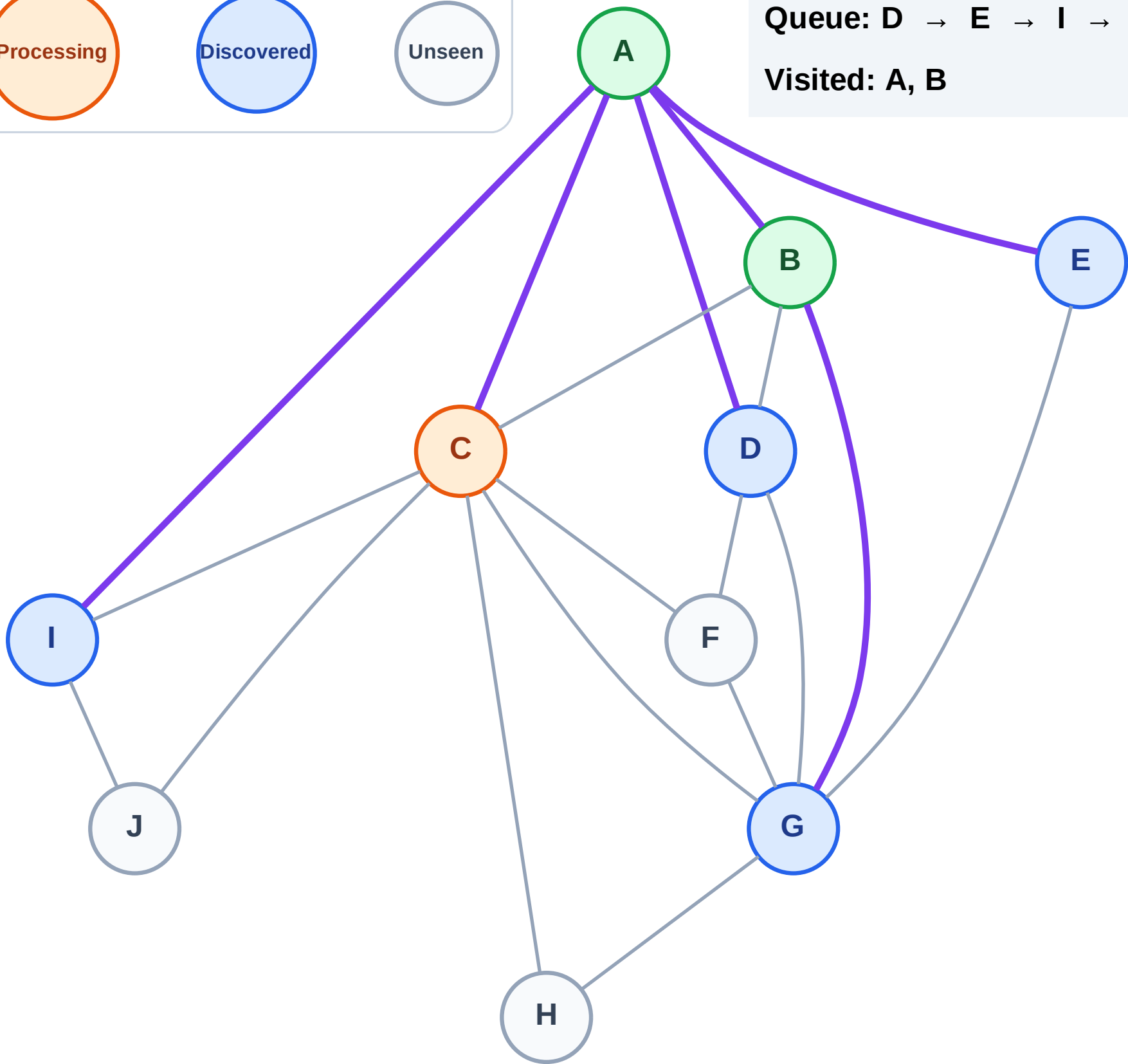
Processing

Discovered

Unseen

Queue: D → E → I → G

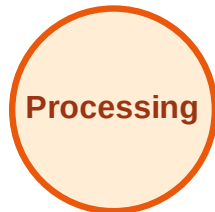
Visited: A, B



BFS Traversal

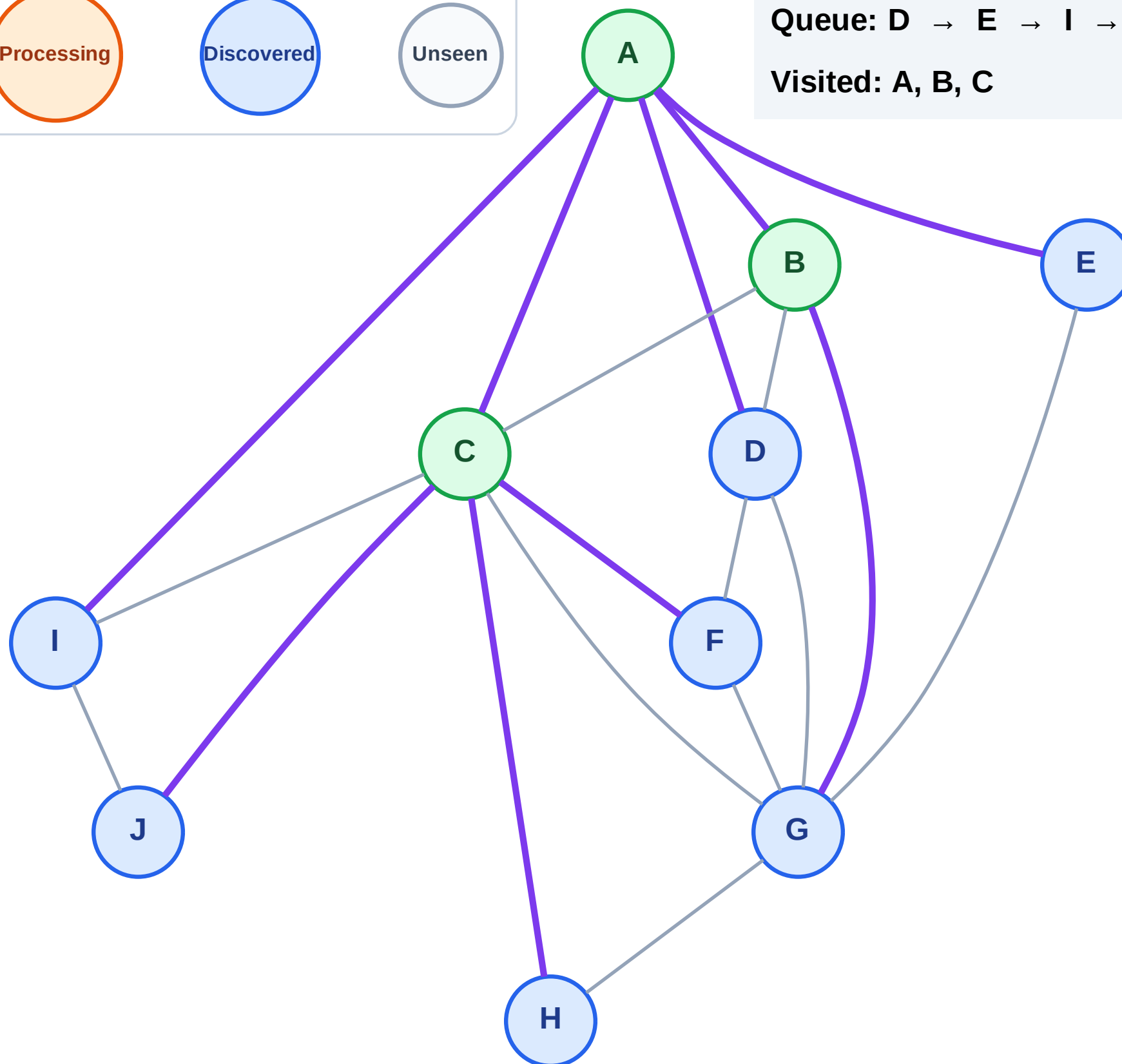
Step 7: Discover F, H, J from C

Legend



Queue: D → E → I → G → F → H → J

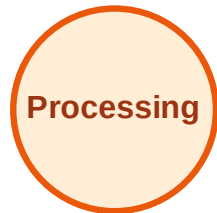
Visited: A, B, C



BFS Traversal

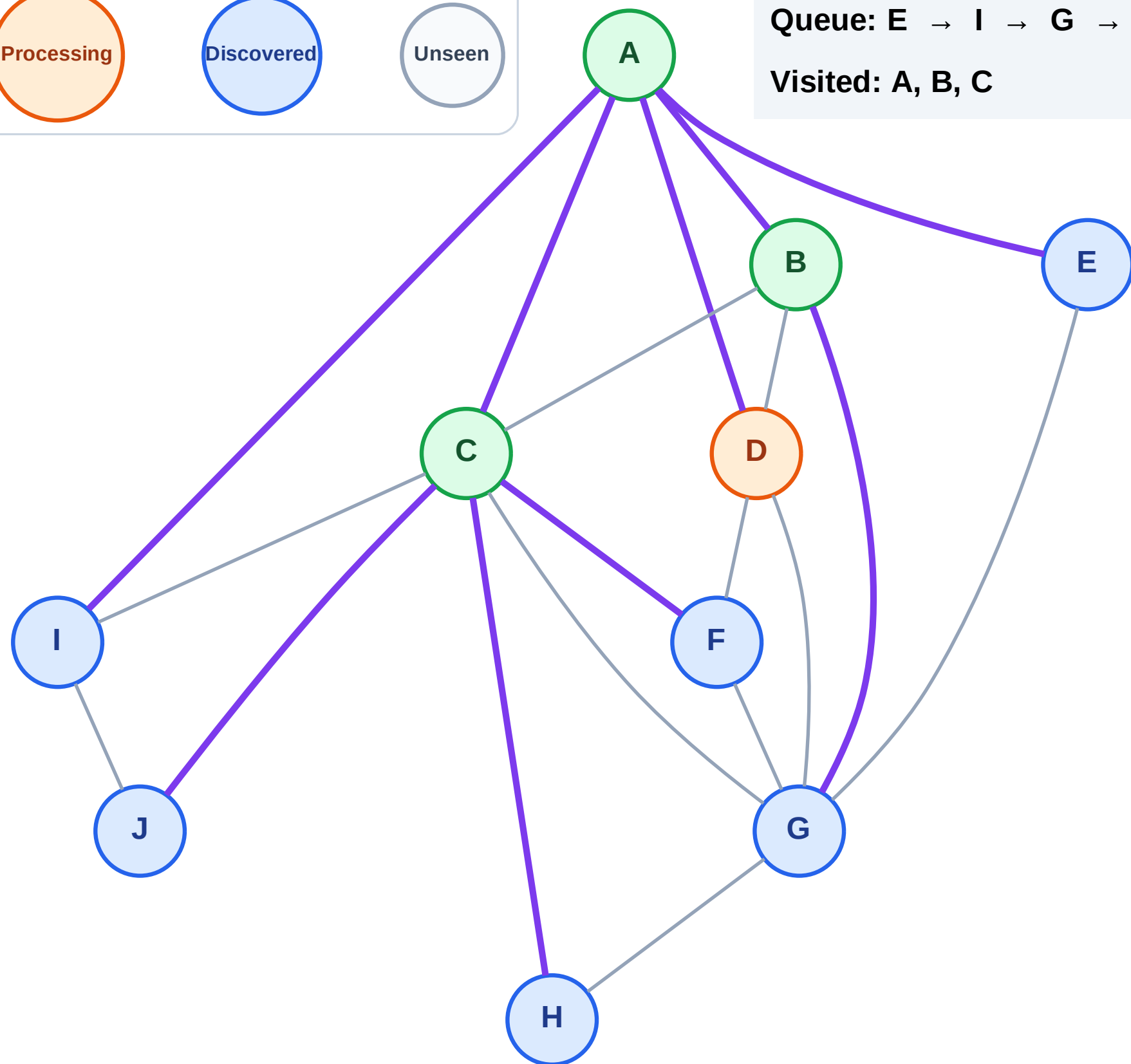
Step 8: Process node D

Legend



Queue: E → I → G → F → H → J

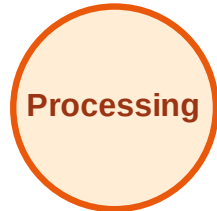
Visited: A, B, C



BFS Traversal

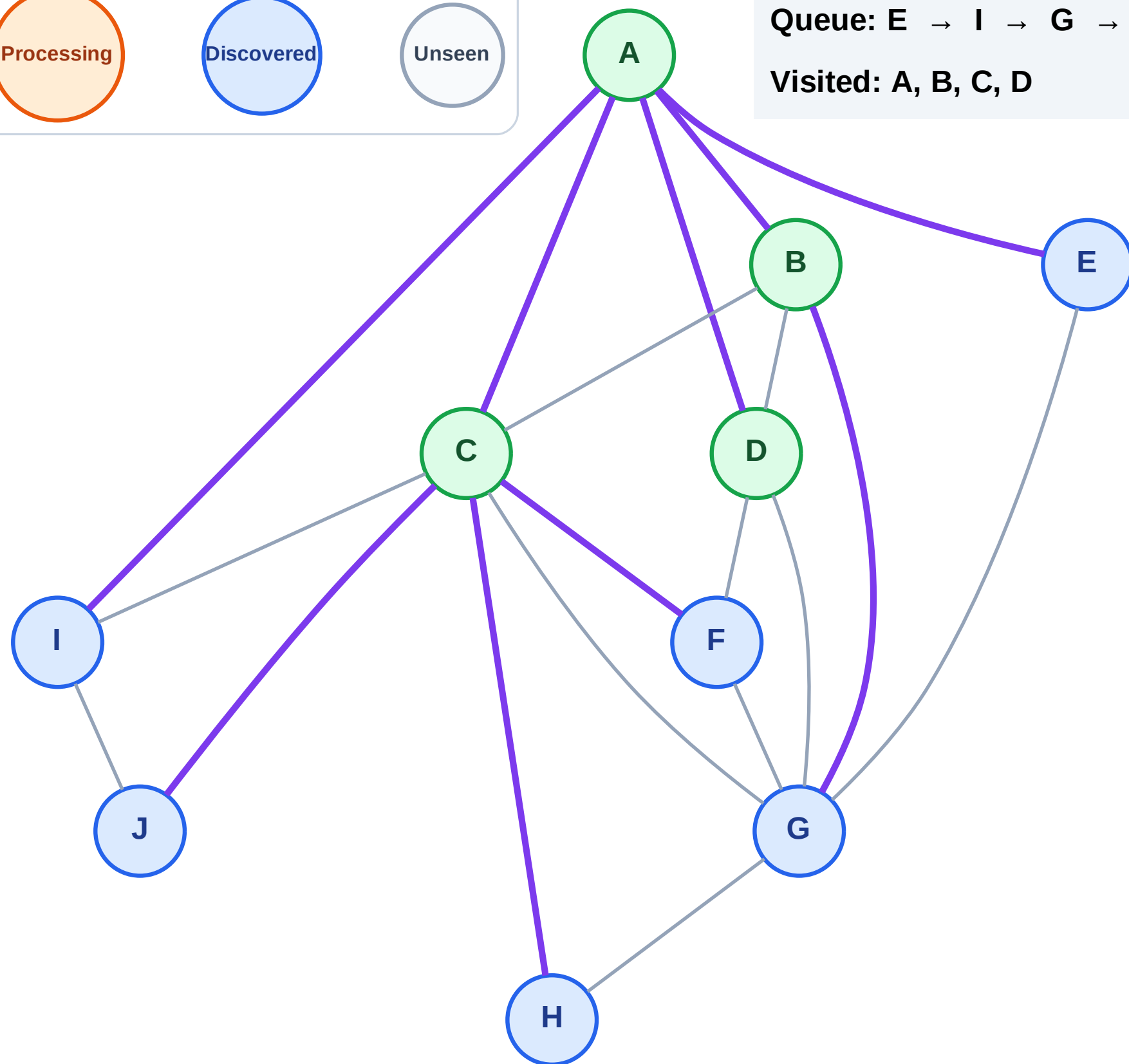
Step 9: No new neighbors from D

Legend



Queue: E → I → G → F → H → J

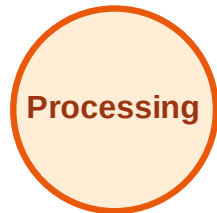
Visited: A, B, C, D



BFS Traversal

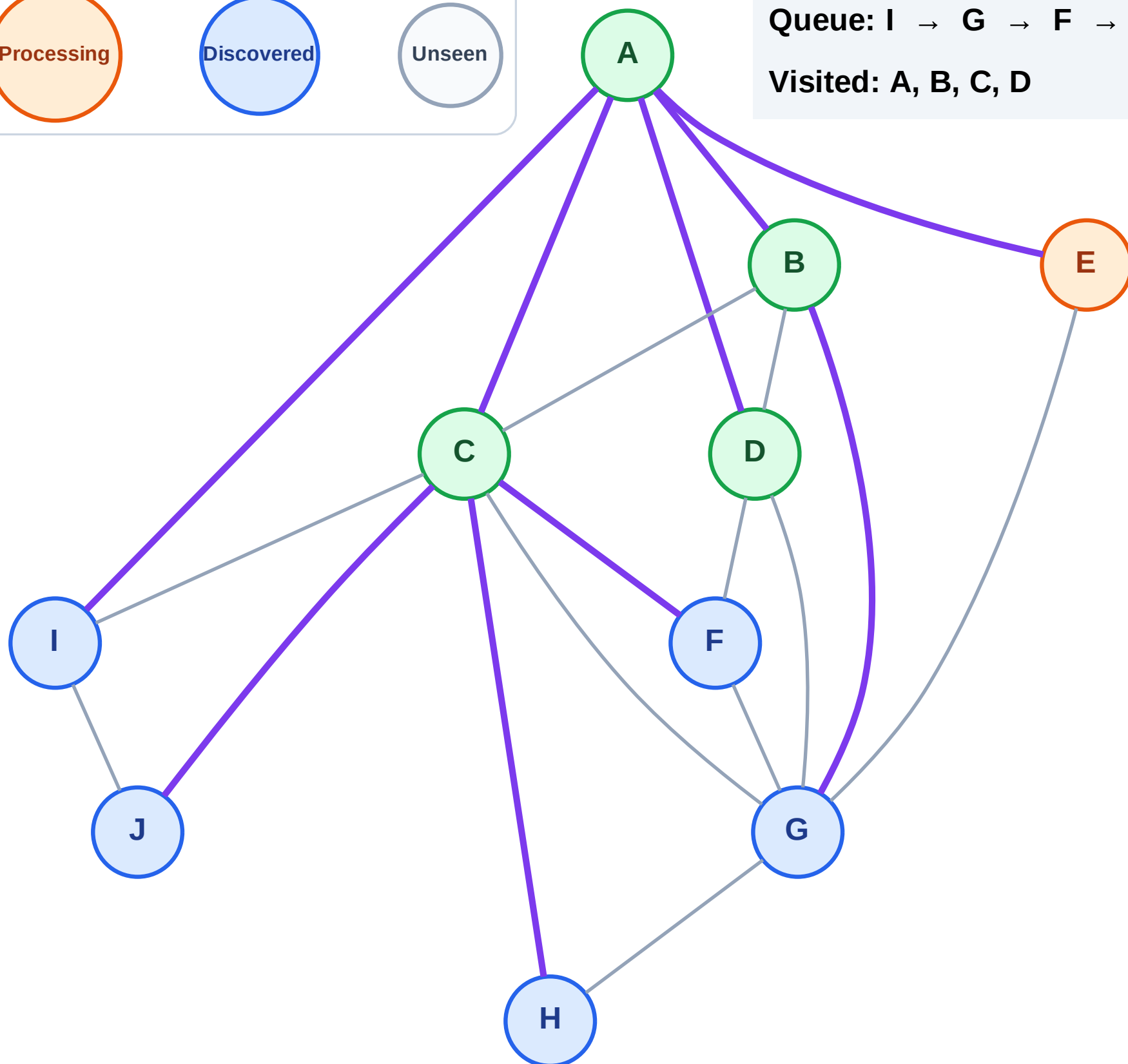
Step 10: Process node E

Legend



Queue: I → G → F → H → J

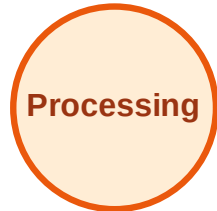
Visited: A, B, C, D



BFS Traversal

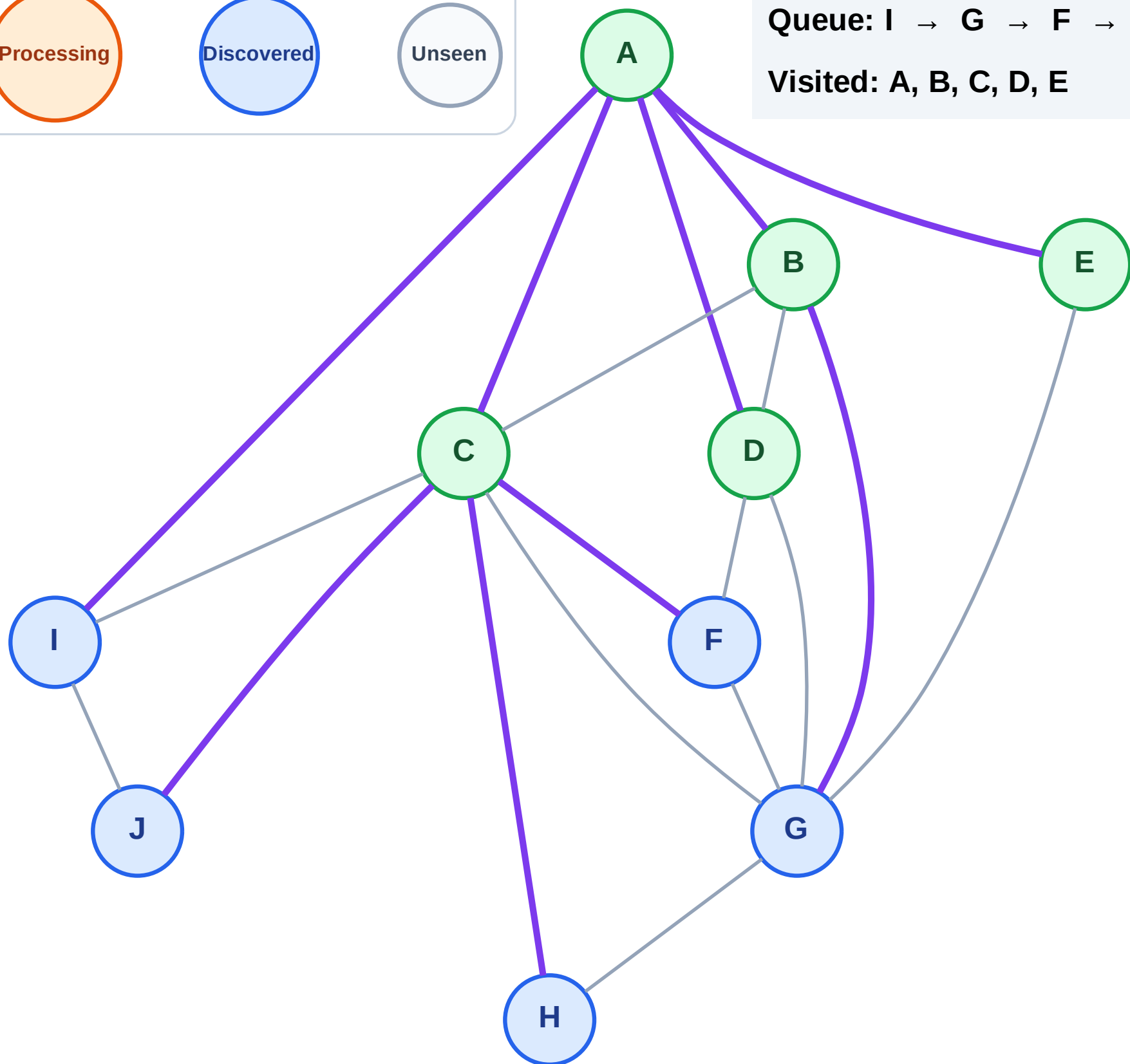
Step 11: No new neighbors from E

Legend



Queue: I → G → F → H → J

Visited: A, B, C, D, E



BFS Traversal

Step 12: Process node I

Legend

Complete

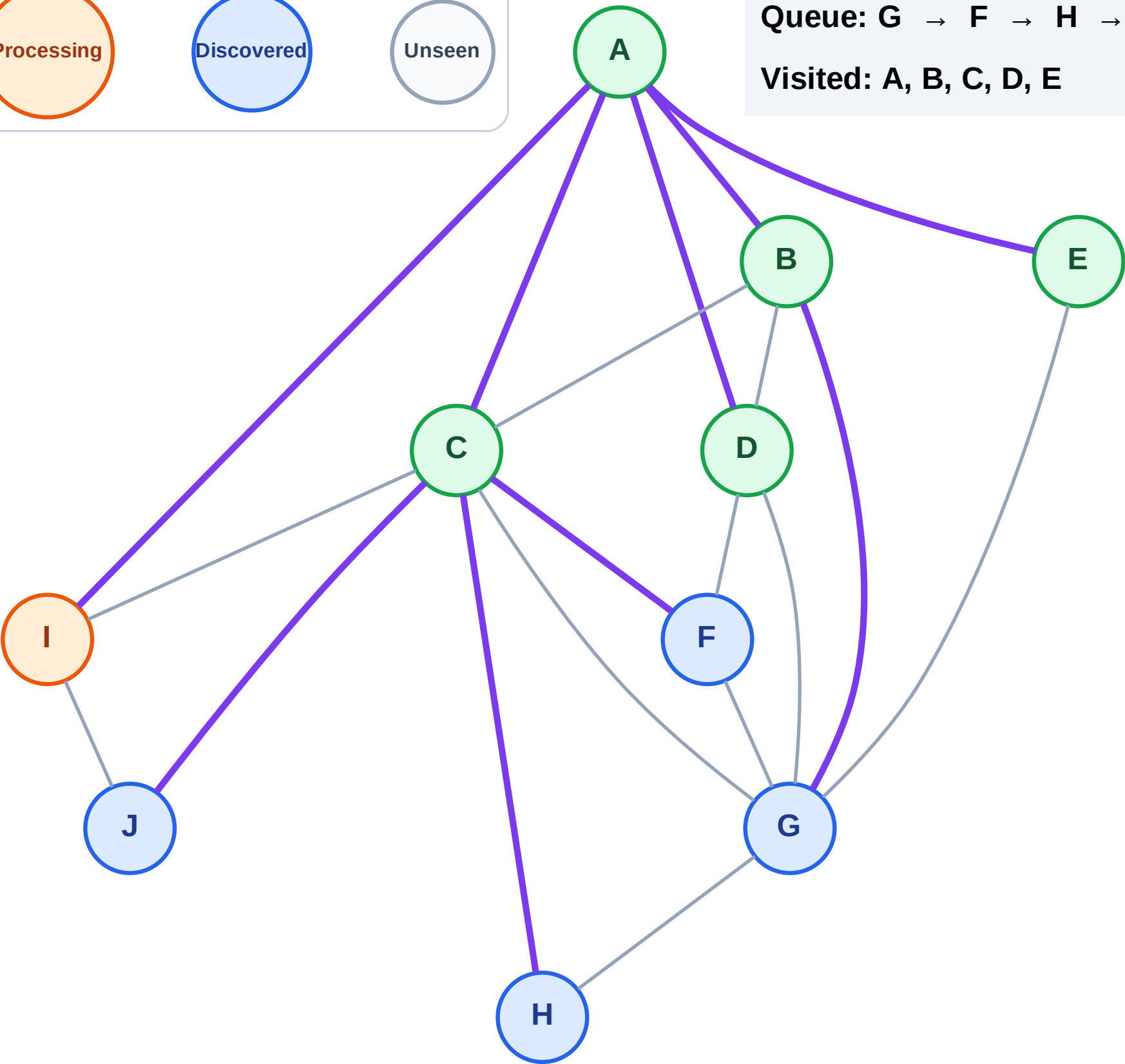
Processing

Discovered

Unseen

Queue: G → F → H → J

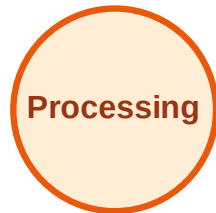
Visited: A, B, C, D, E



BFS Traversal

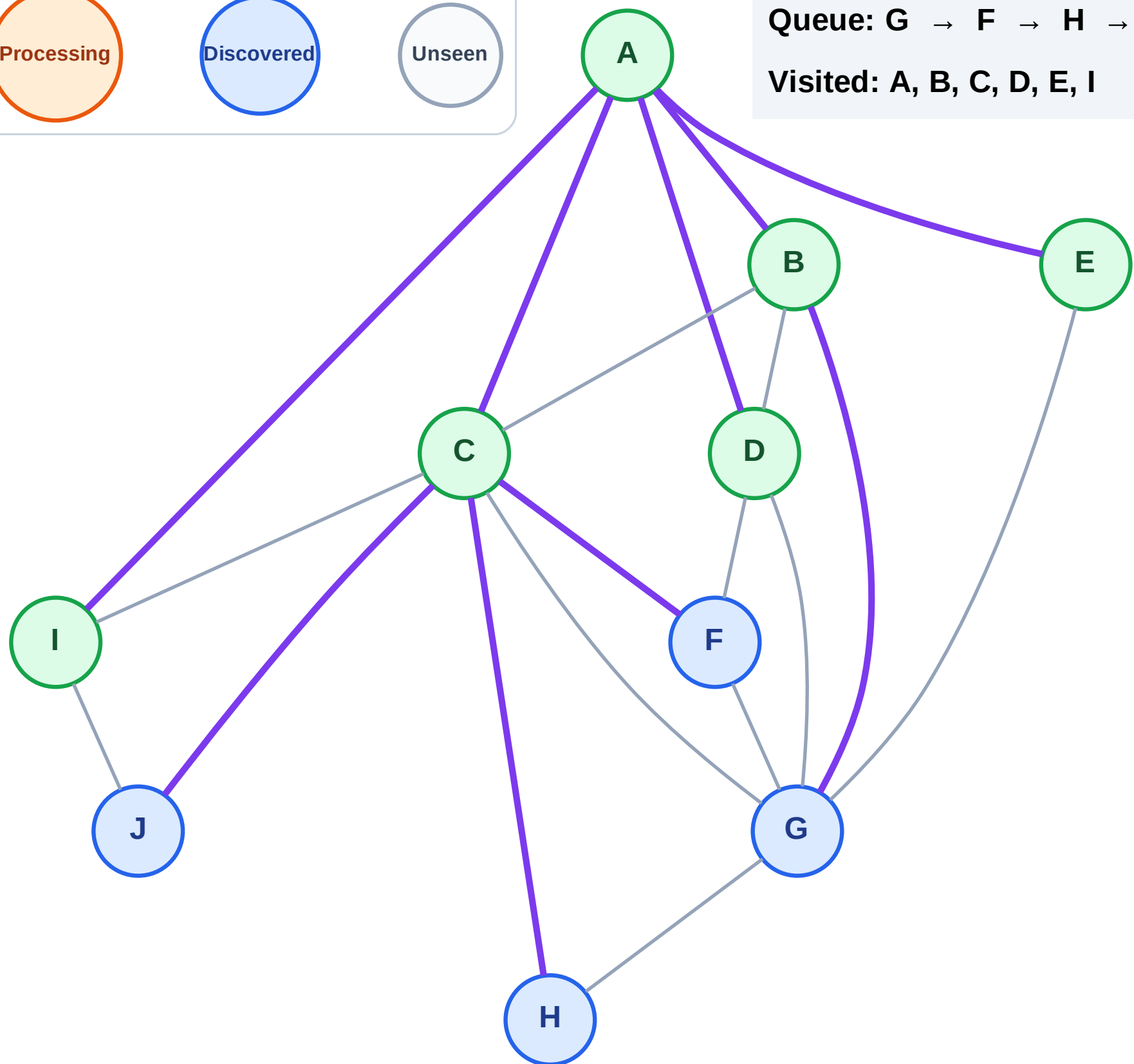
Step 13: No new neighbors from I

Legend



Queue: G → F → H → J

Visited: A, B, C, D, E, I



BFS Traversal

Step 14: Process node G

Legend

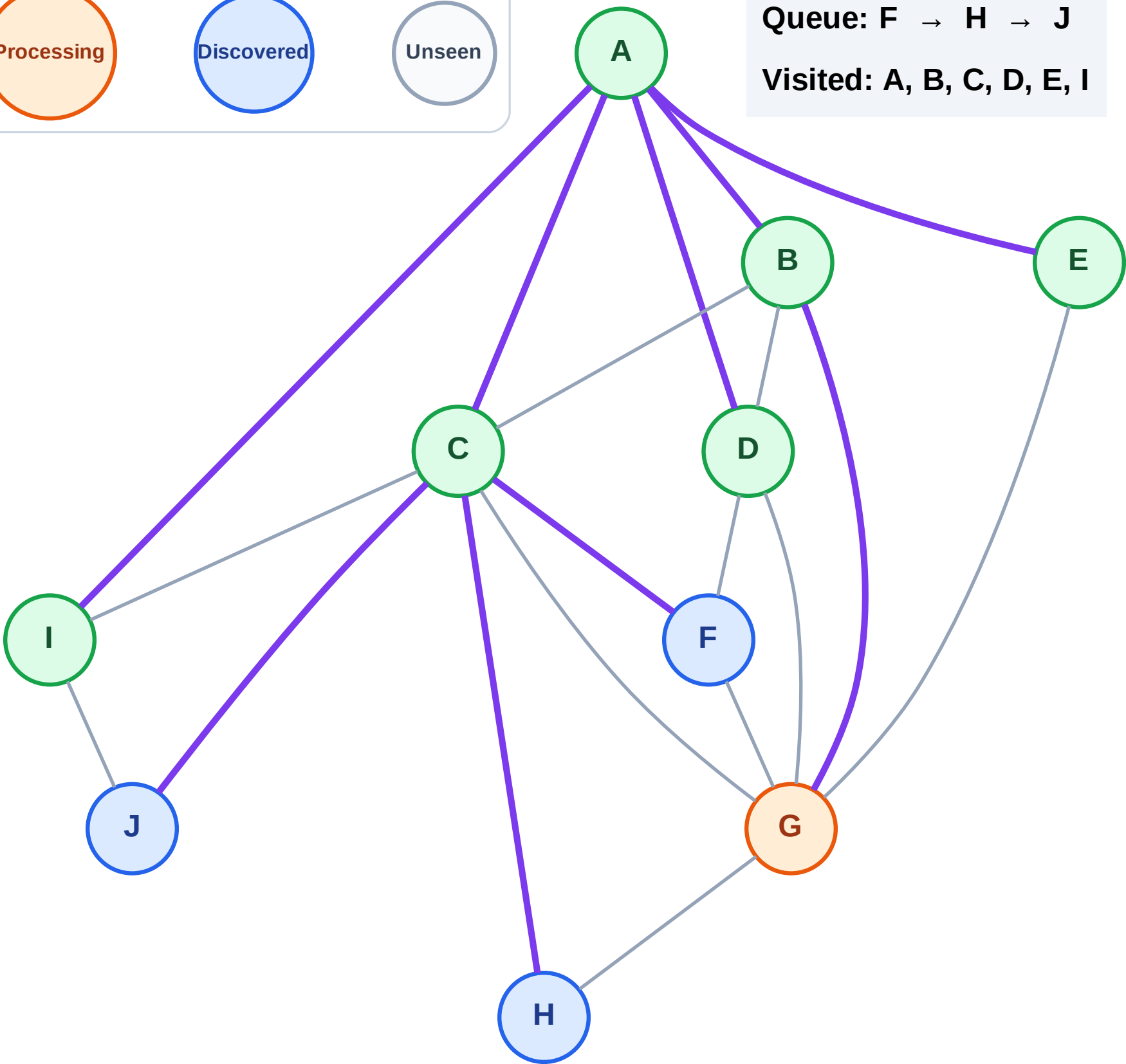
Complete

Processing

Discovered

Unseen

Queue: F → H → J
Visited: A, B, C, D, E, I



BFS Traversal

Step 15: No new neighbors from G

Legend

Complete

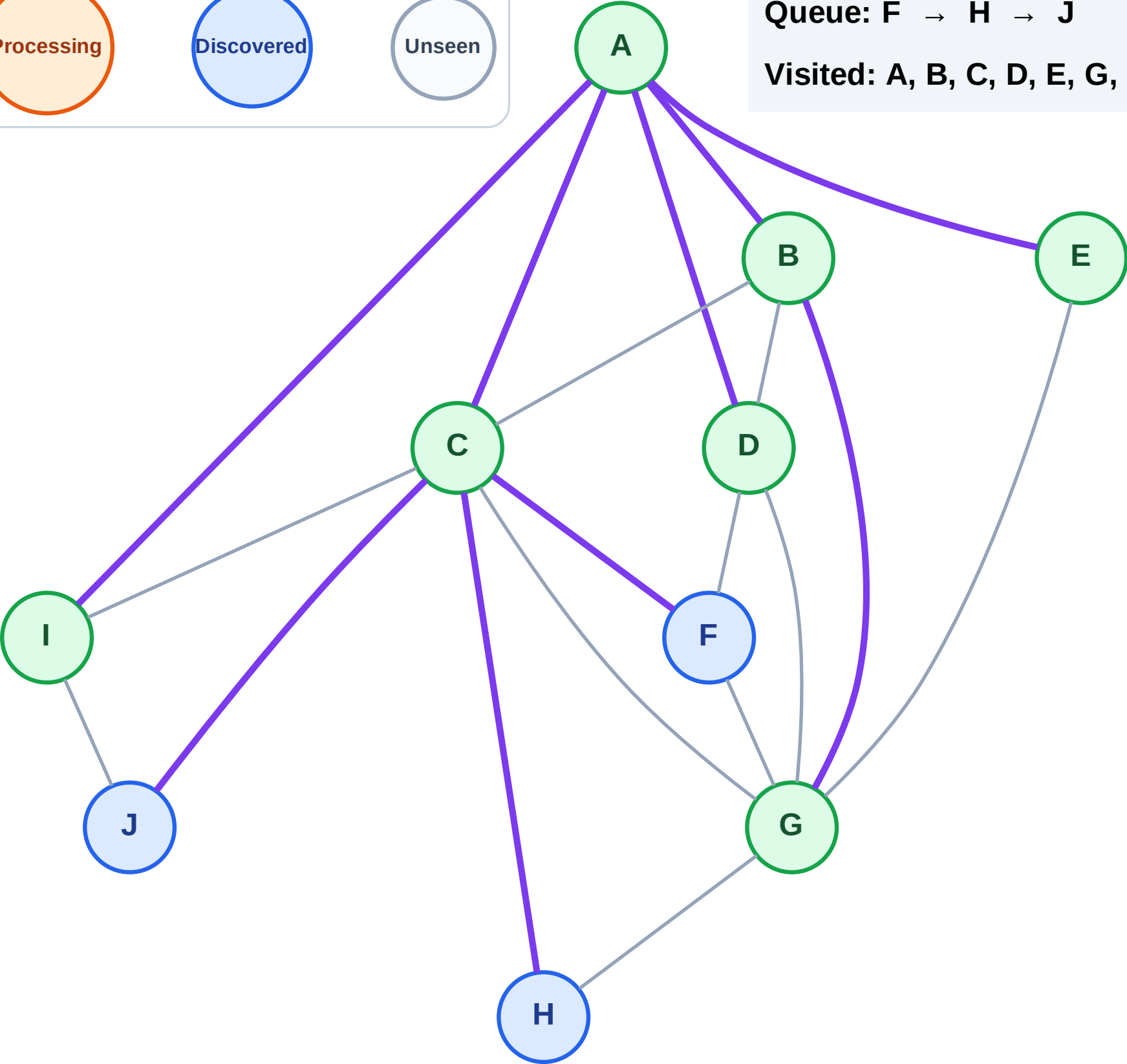
Processing

Discovered

Unseen

Queue: F → H → J

Visited: A, B, C, D, E, G, I



BFS Traversal

Step 16: Process node F

Legend

Complete

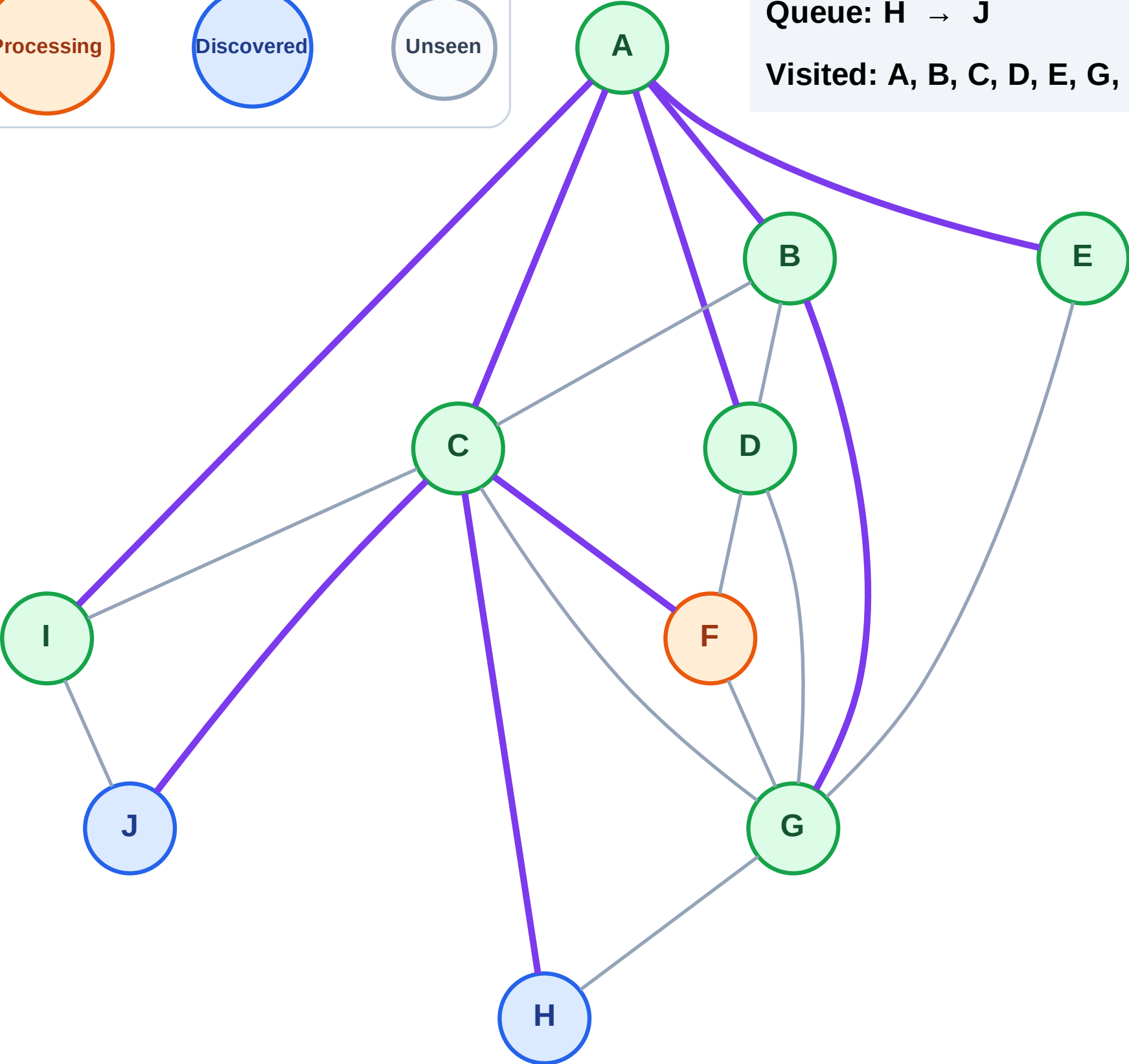
Processing

Discovered

Unseen

Queue: H → J

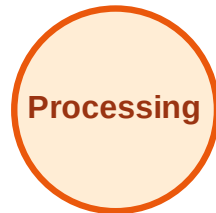
Visited: A, B, C, D, E, G, I



BFS Traversal

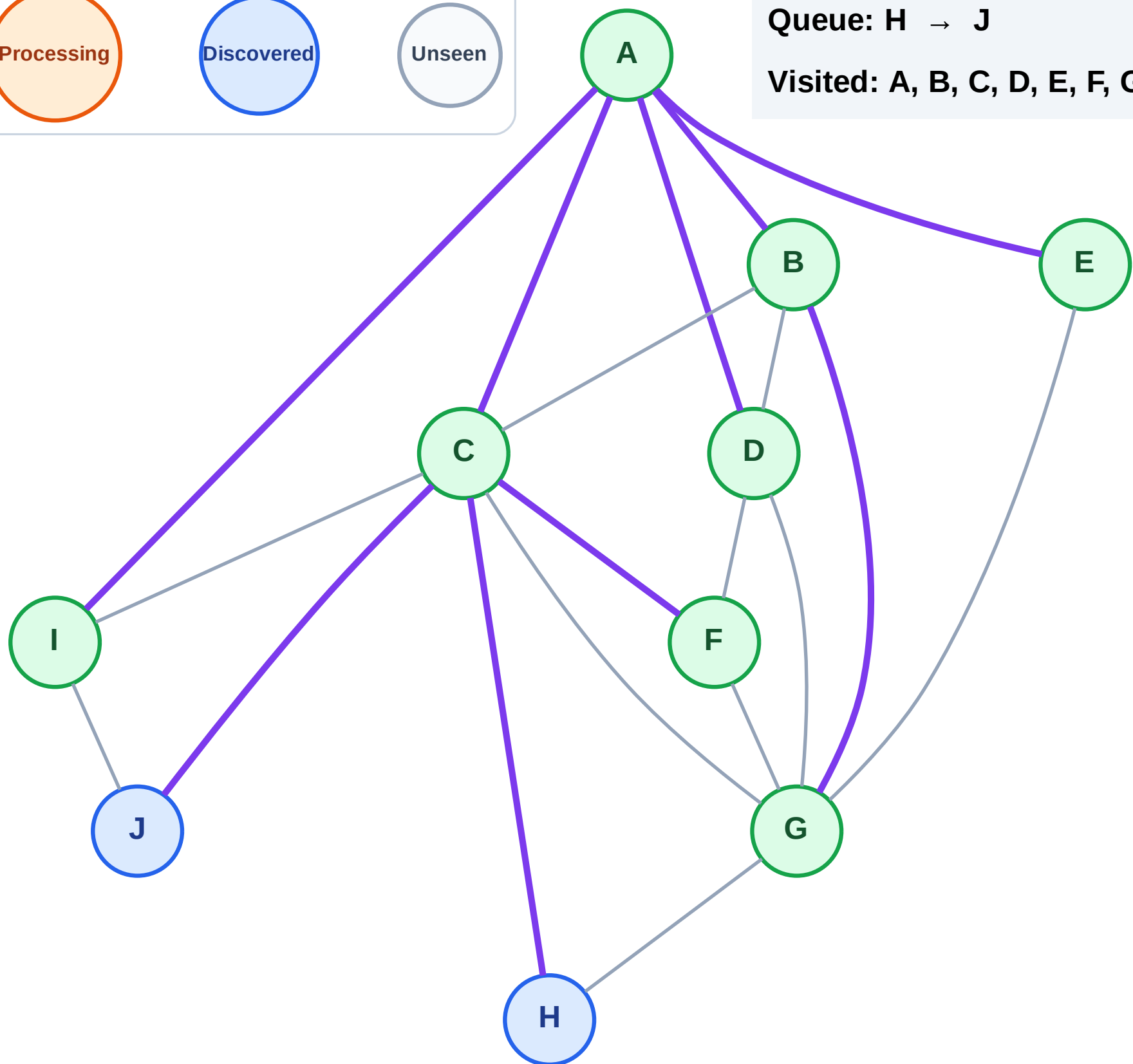
Step 17: No new neighbors from F

Legend



Queue: H → J

Visited: A, B, C, D, E, F, G, I



BFS Traversal

Step 18: Process node H

Legend

Complete

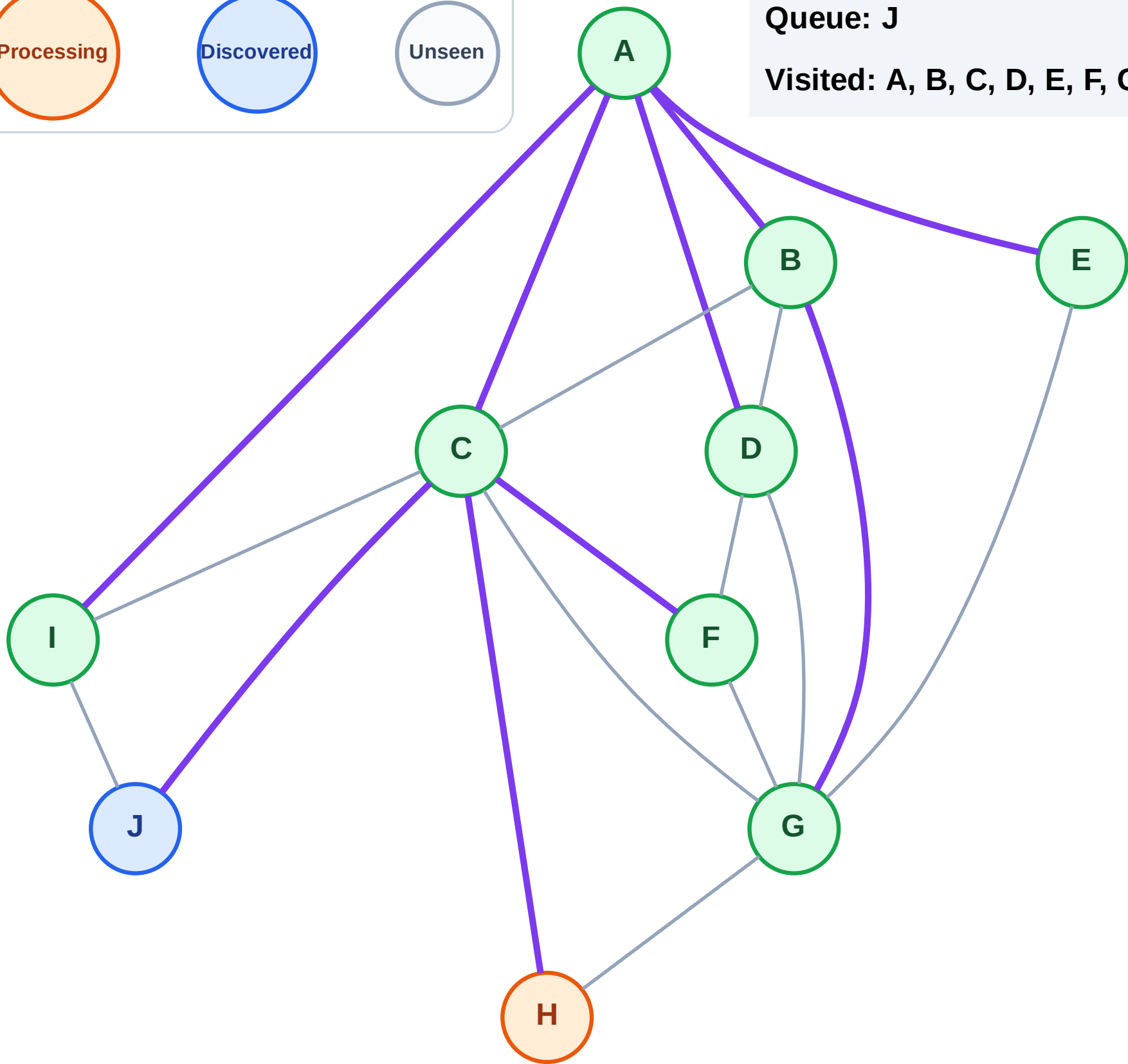
Processing

Discovered

Unseen

Queue: J

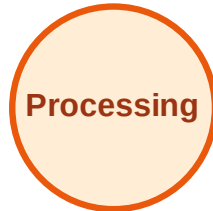
Visited: A, B, C, D, E, F, G, I



BFS Traversal

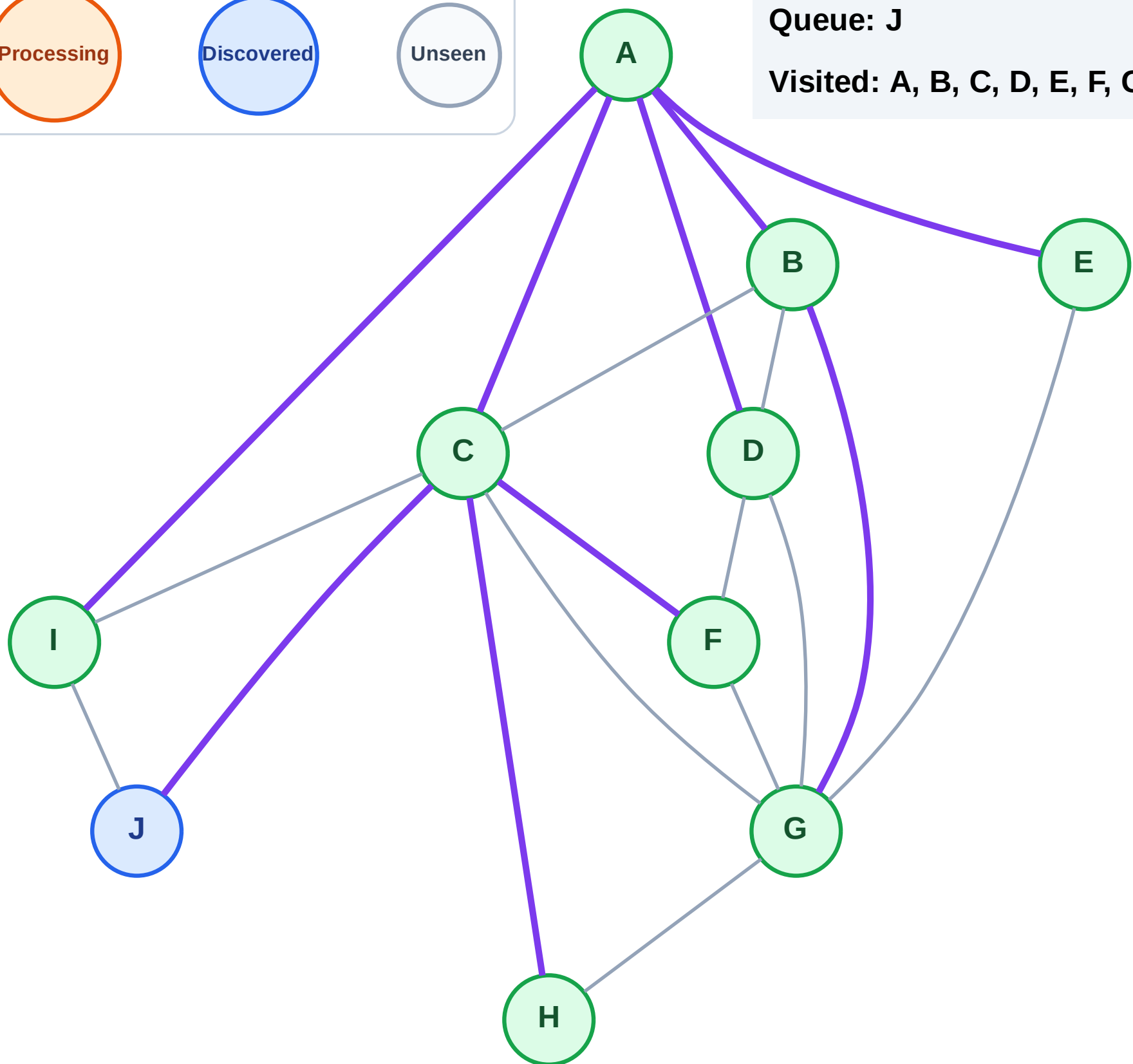
Step 19: No new neighbors from H

Legend



Queue: J

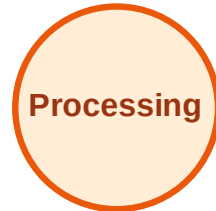
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

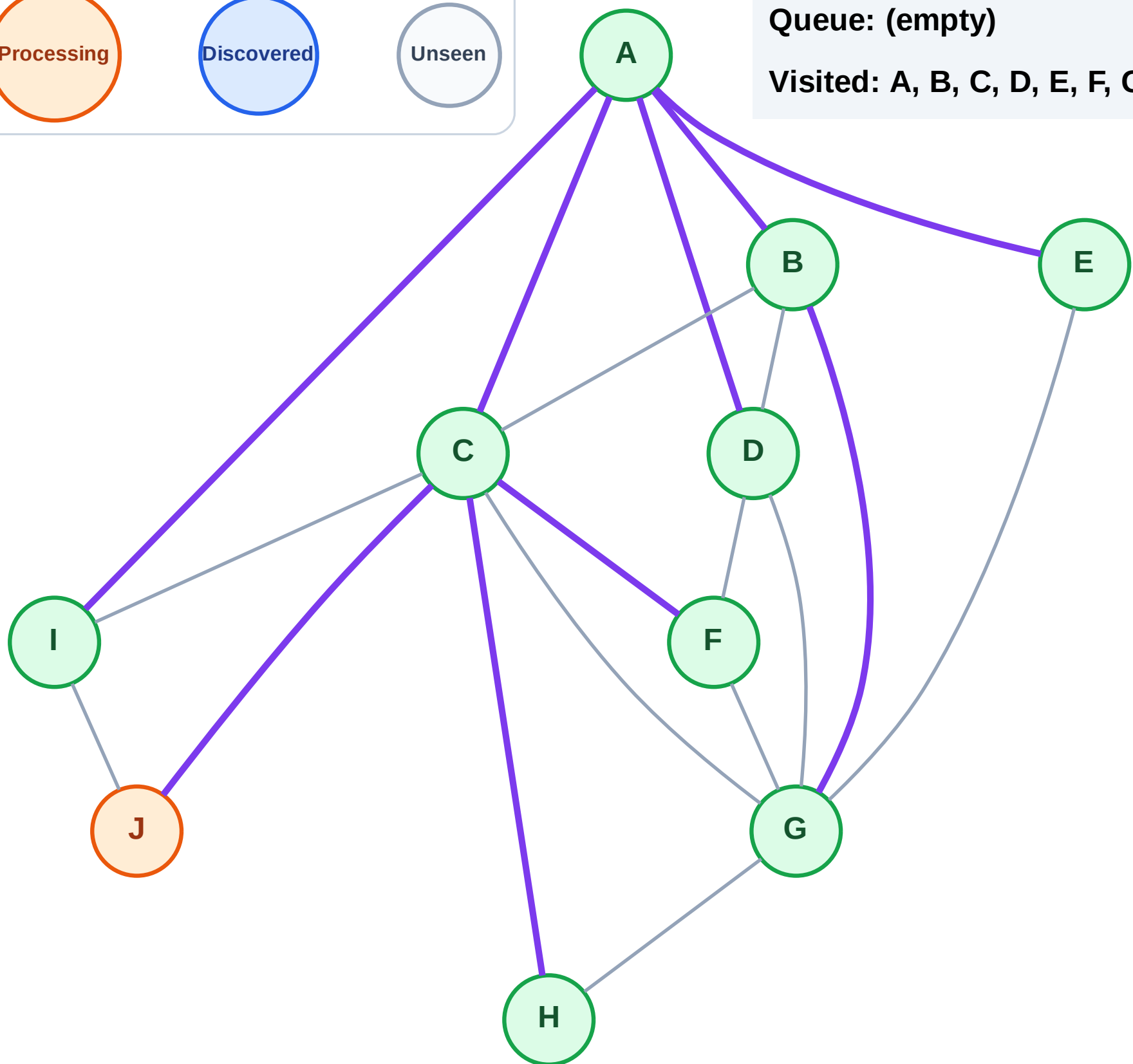
Step 20: Process node J

Legend



Queue: (empty)

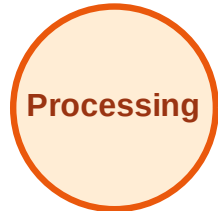
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

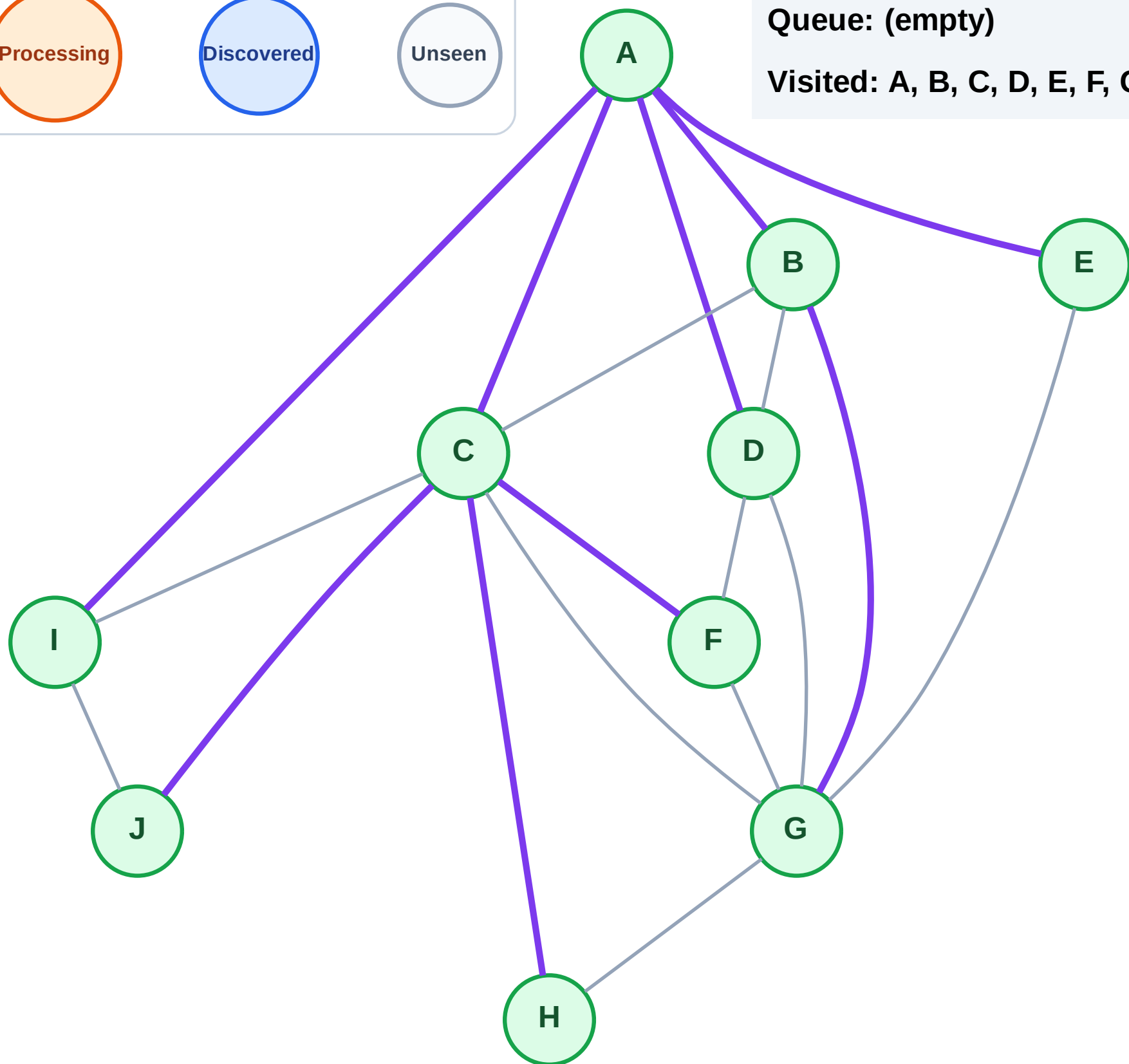
Step 21: No new neighbors from J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

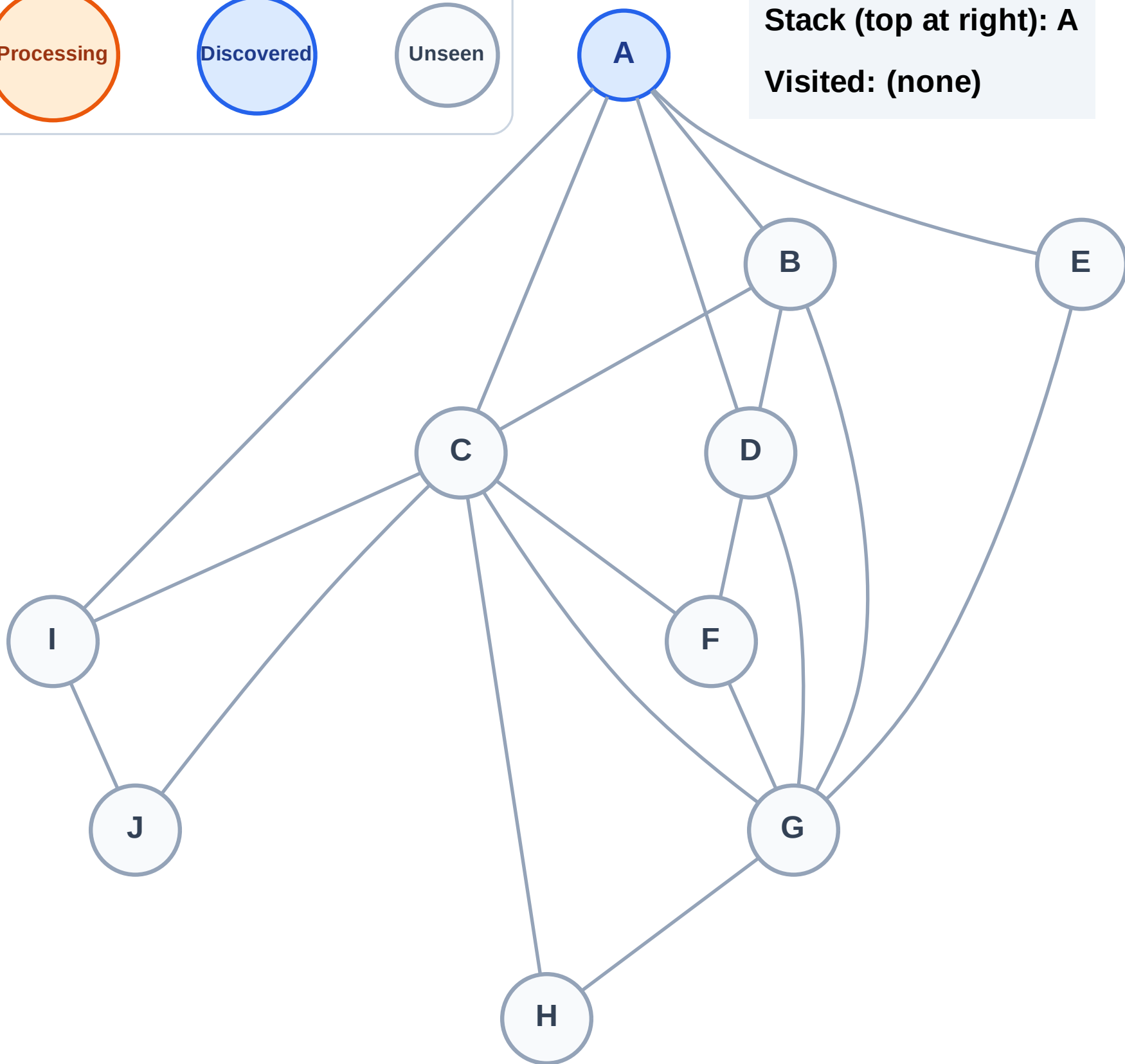
Step 1: Start at A

Legend



Stack (top at right): A

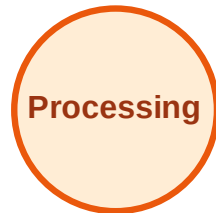
Visited: (none)



DFS Traversal

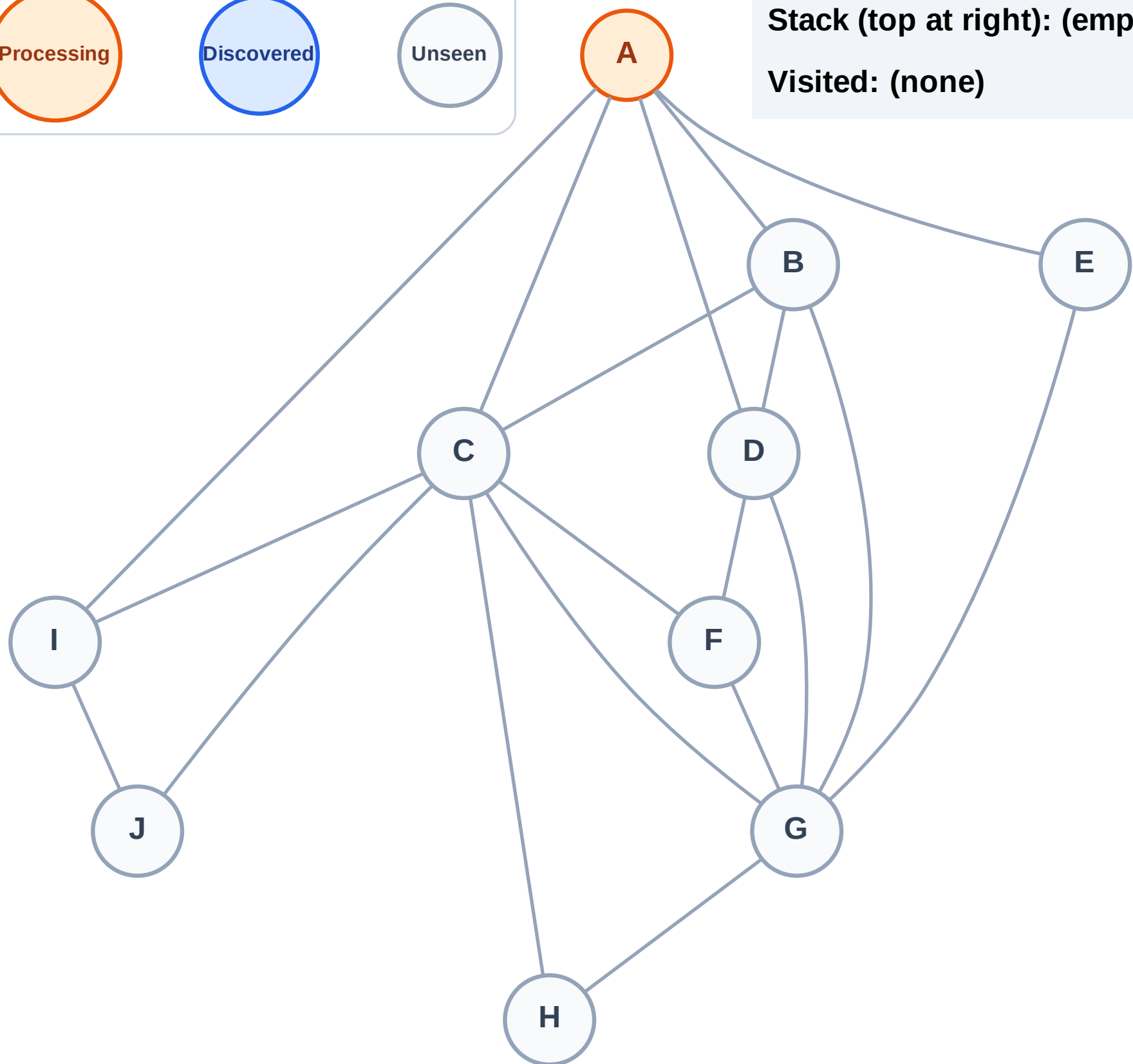
Step 2: Process node A

Legend



Stack (top at right): (empty)

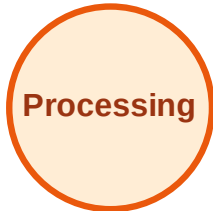
Visited: (none)



DFS Traversal

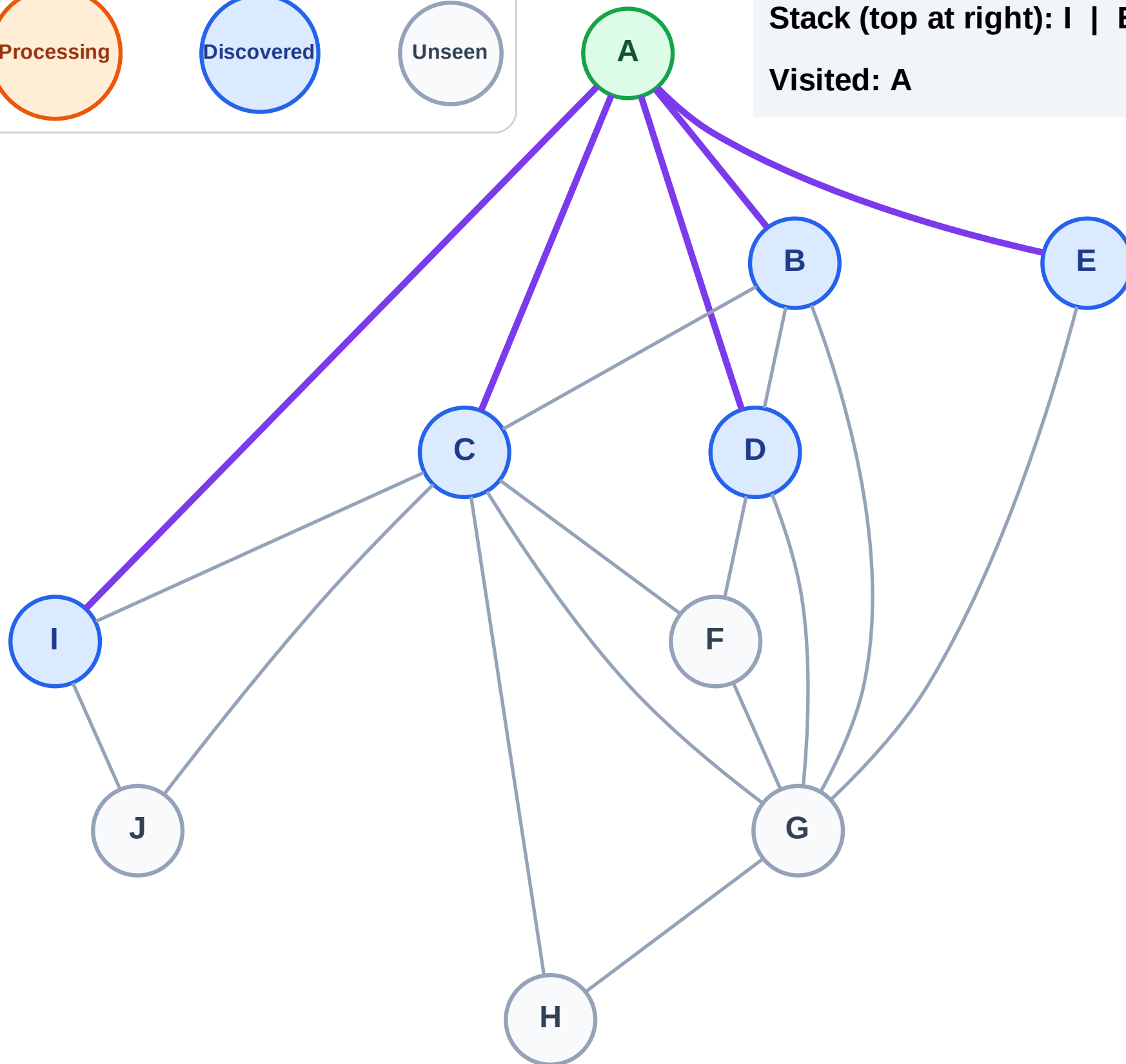
Step 3: Discover I, E, D, C, B from A

Legend



Stack (top at right): I | E | D | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

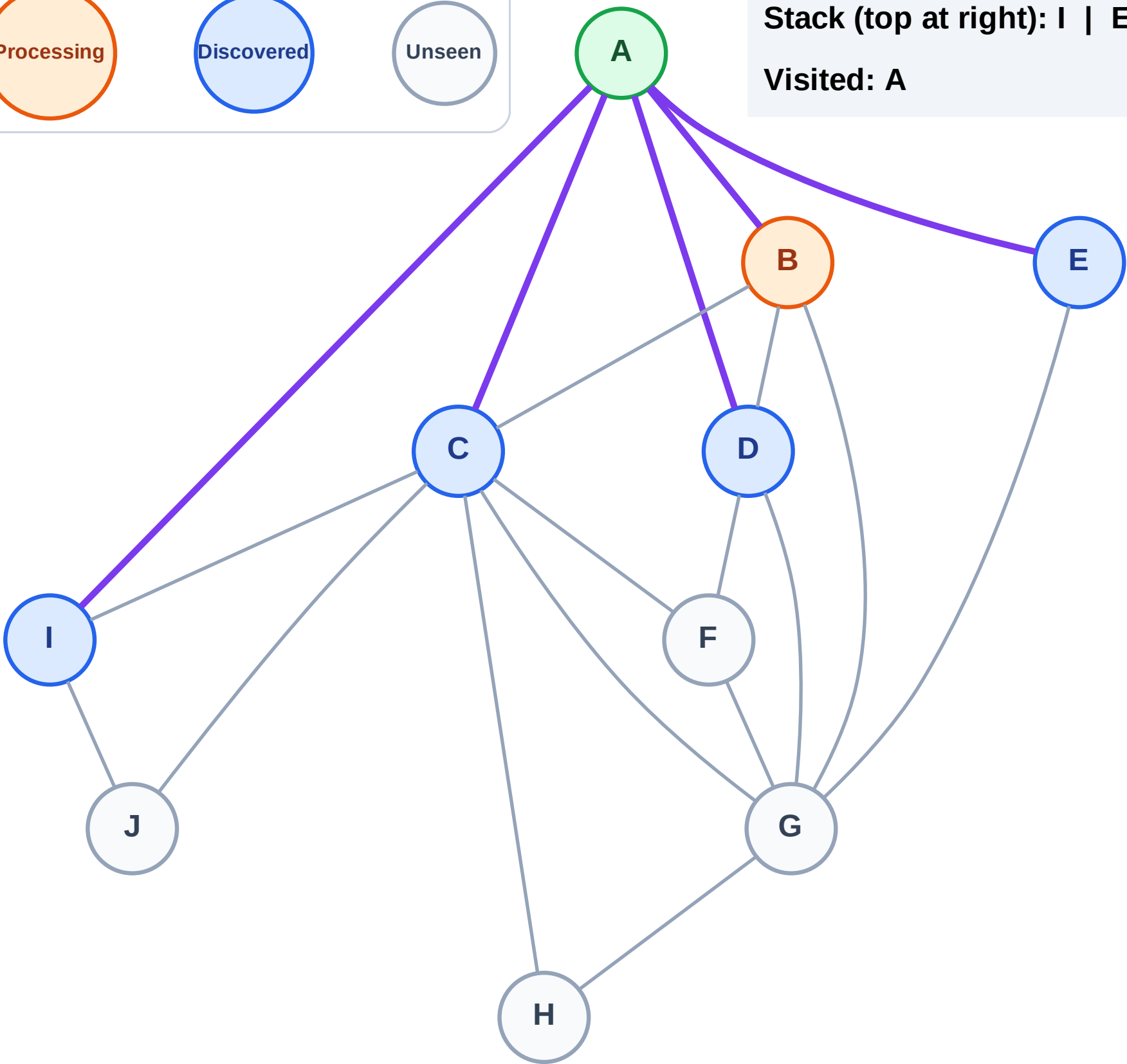
Complete

Processing

Discovered

Unseen

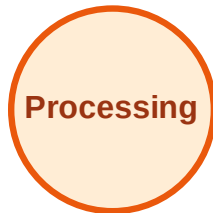
Stack (top at right): I | E | D | C
Visited: A



DFS Traversal

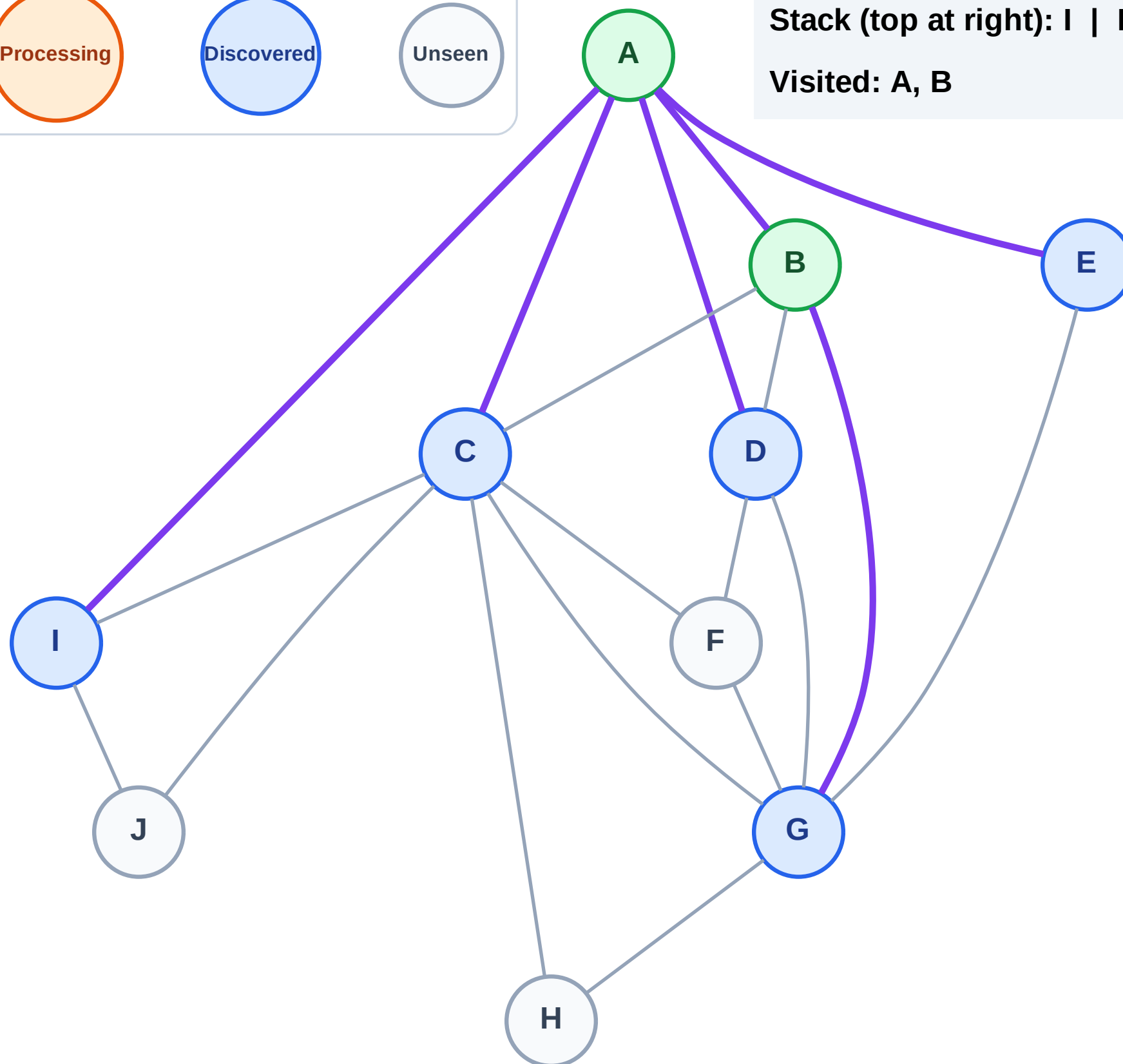
Step 5: Discover G from B

Legend



Stack (top at right): I | E | D | C | G

Visited: A, B



DFS Traversal

Step 6: Process node G

Legend

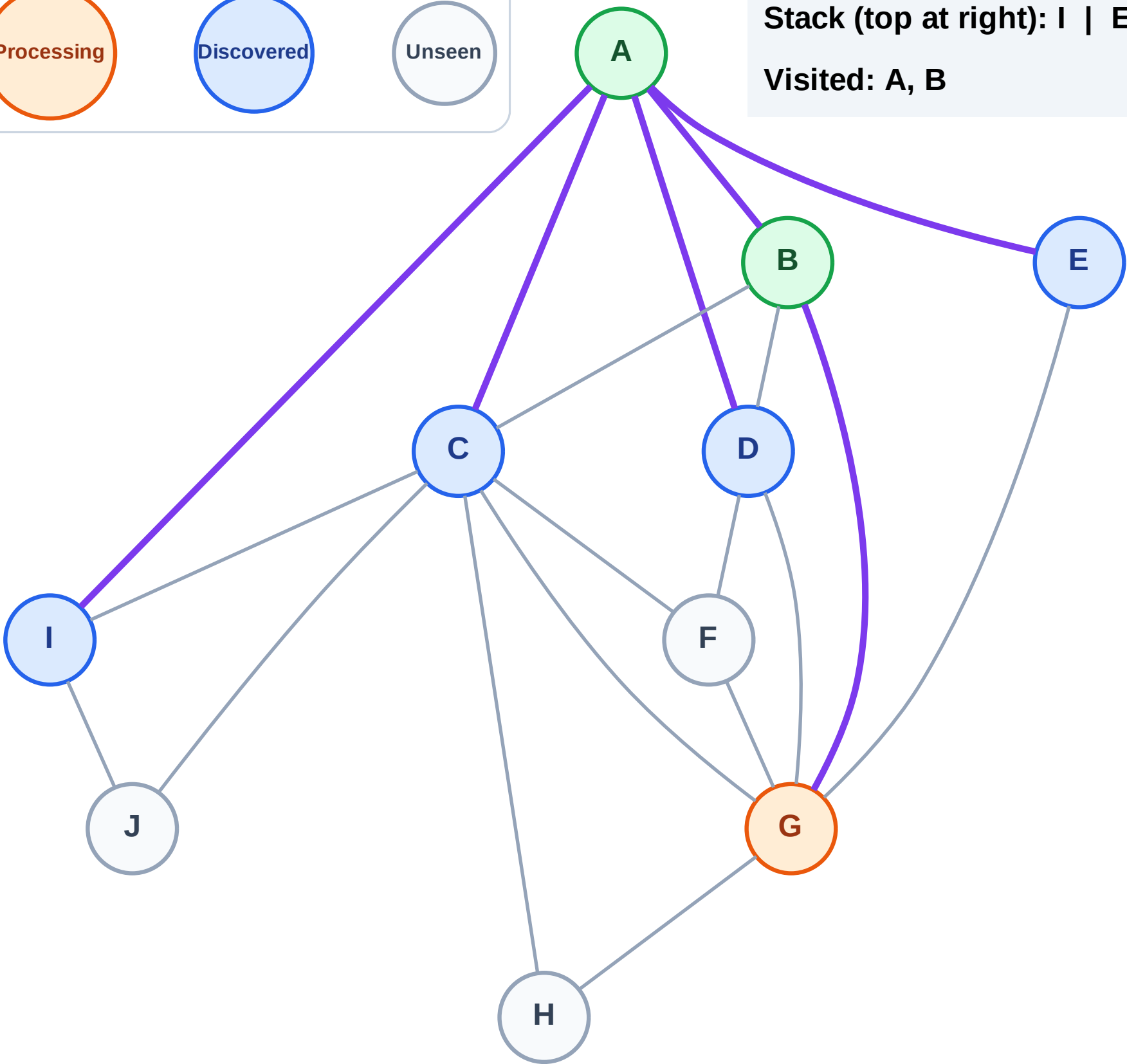
Complete

Processing

Discovered

Unseen

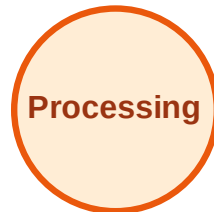
Stack (top at right): I | E | D | C
Visited: A, B



DFS Traversal

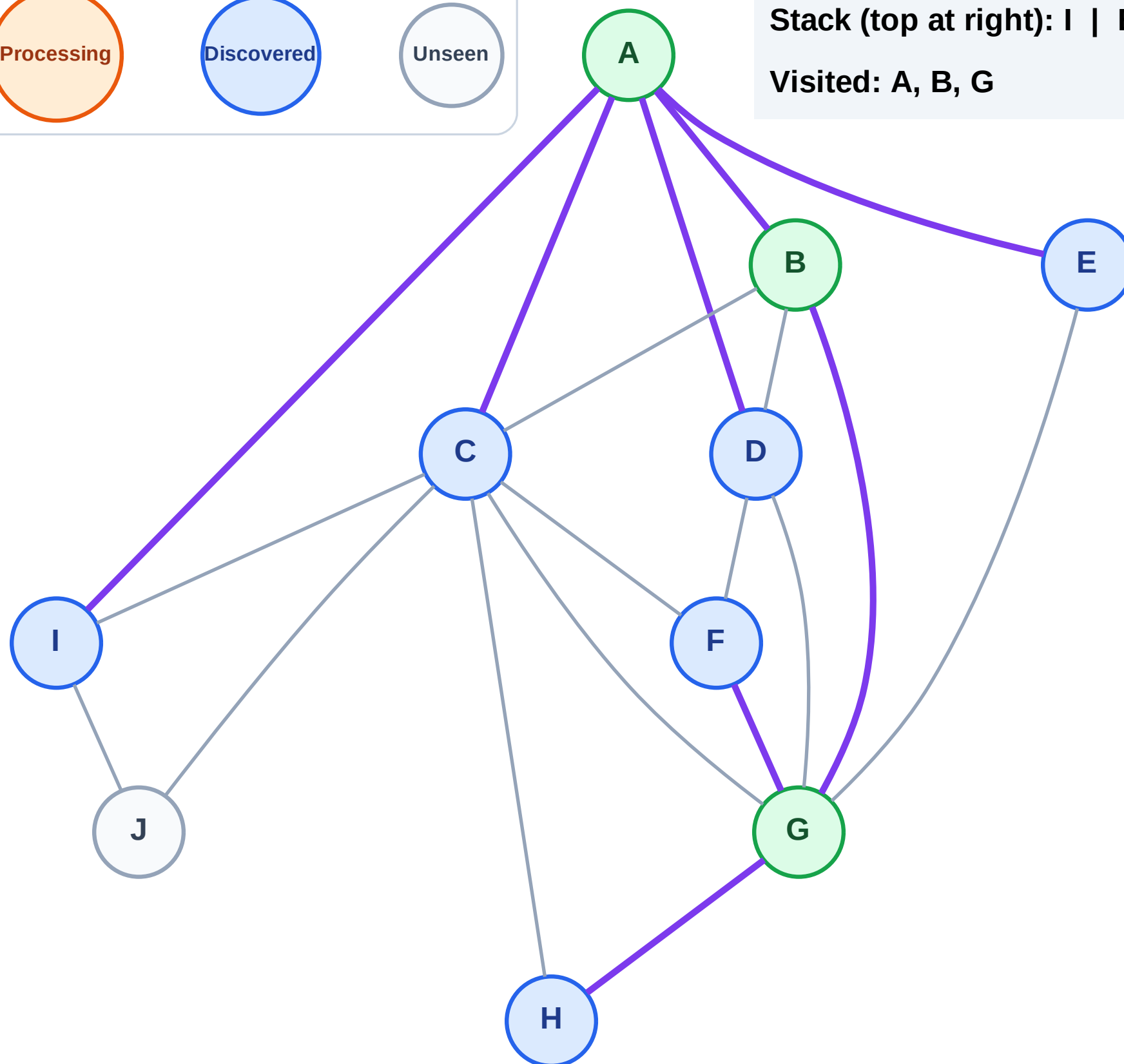
Step 7: Discover H, F from G

Legend



Stack (top at right): I | E | D | C | H | F

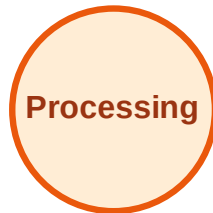
Visited: A, B, G



DFS Traversal

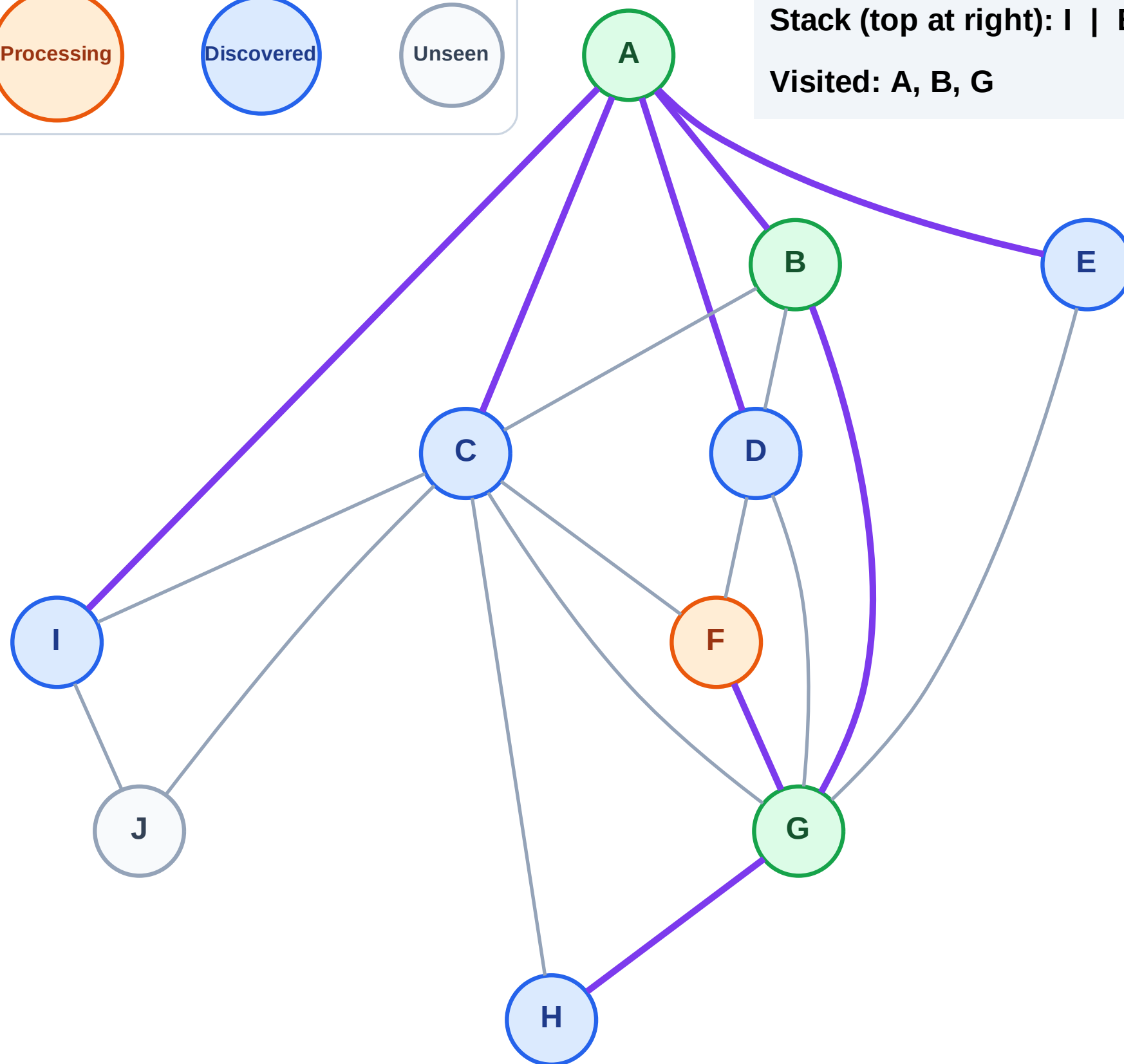
Step 8: Process node F

Legend



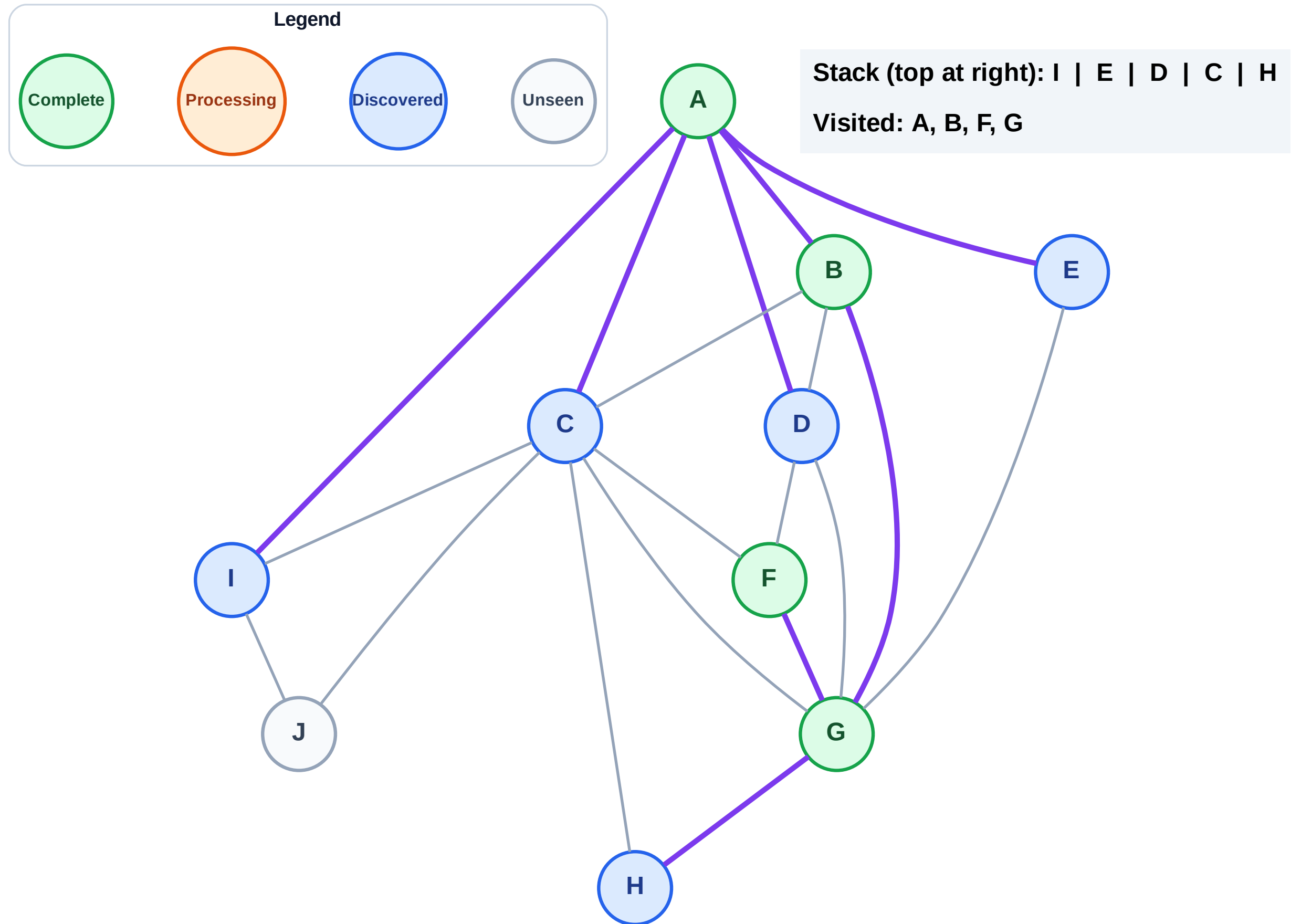
Stack (top at right): I | E | D | C | H

Visited: A, B, G



Step 9: No new neighbors from F

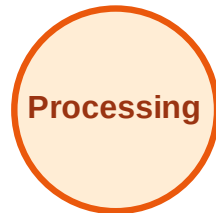
Step 9: No new neighbors from F



DFS Traversal

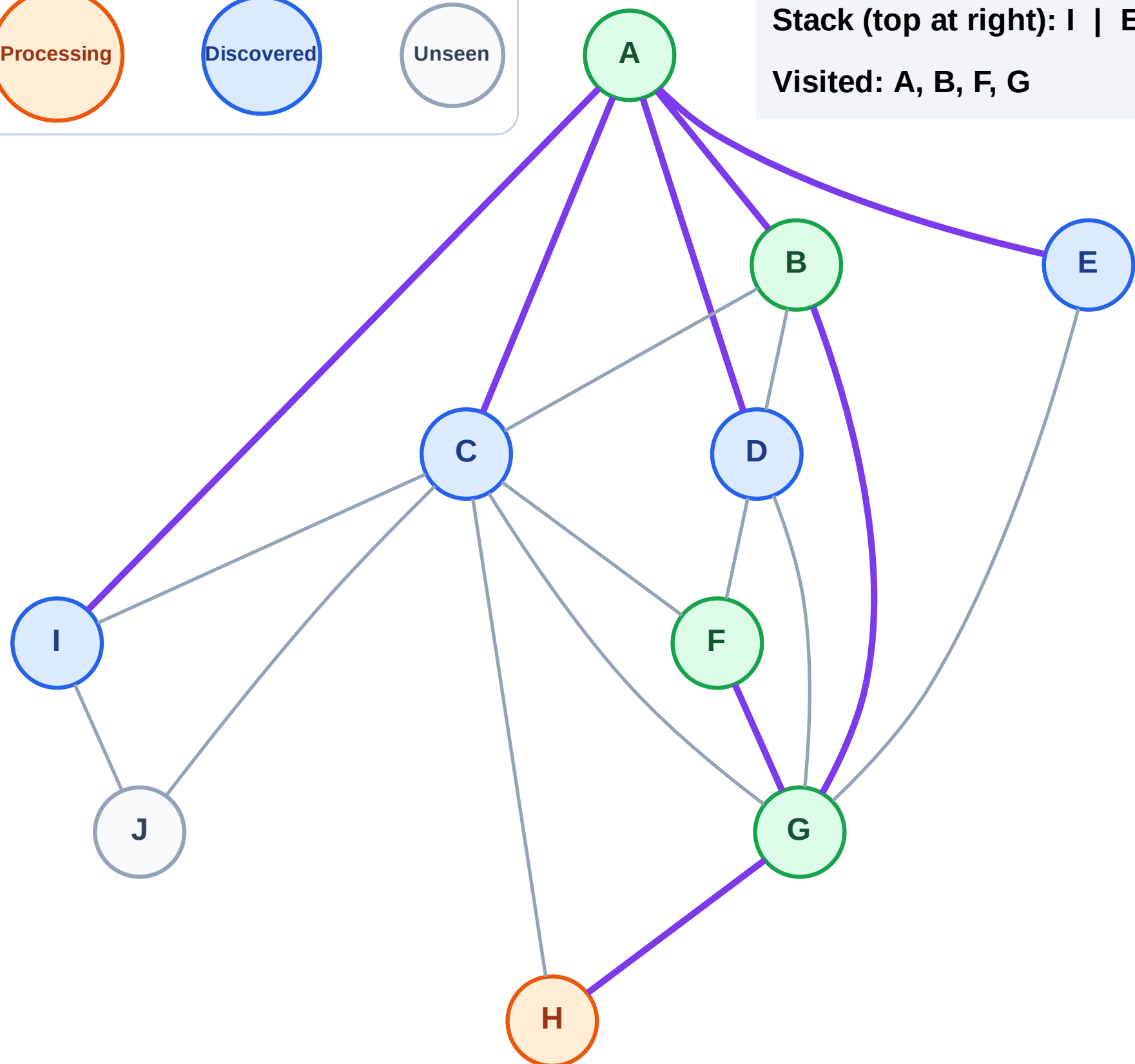
Step 10: Process node H

Legend



Stack (top at right): I | E | D | C

Visited: A, B, F, G



DFS Traversal

Step 11: No new neighbors from H

Legend

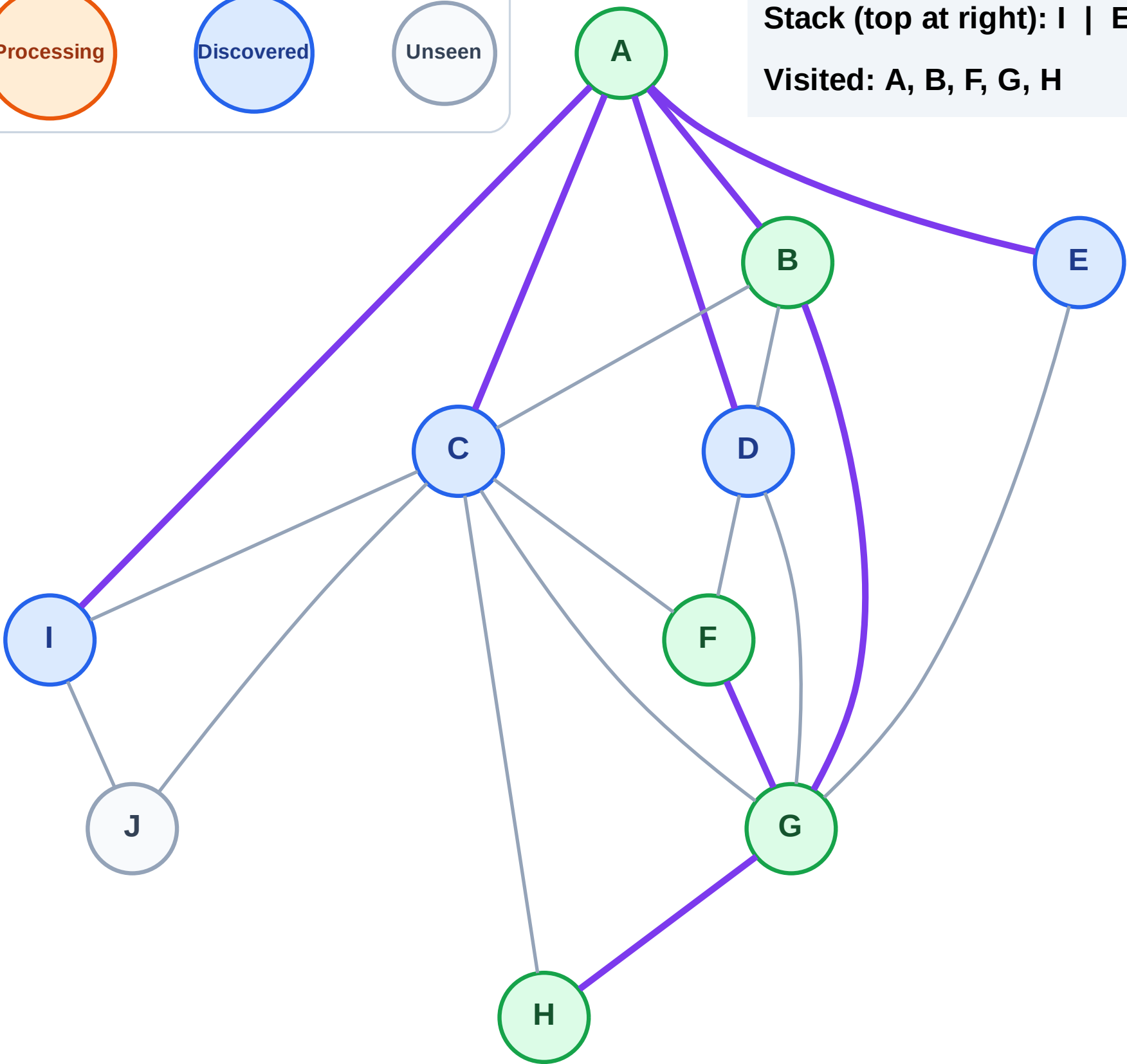
Complete

Processing

Discovered

Unseen

Stack (top at right): I | E | D | C
Visited: A, B, F, G, H



DFS Traversal

Step 12: Process node C

Legend

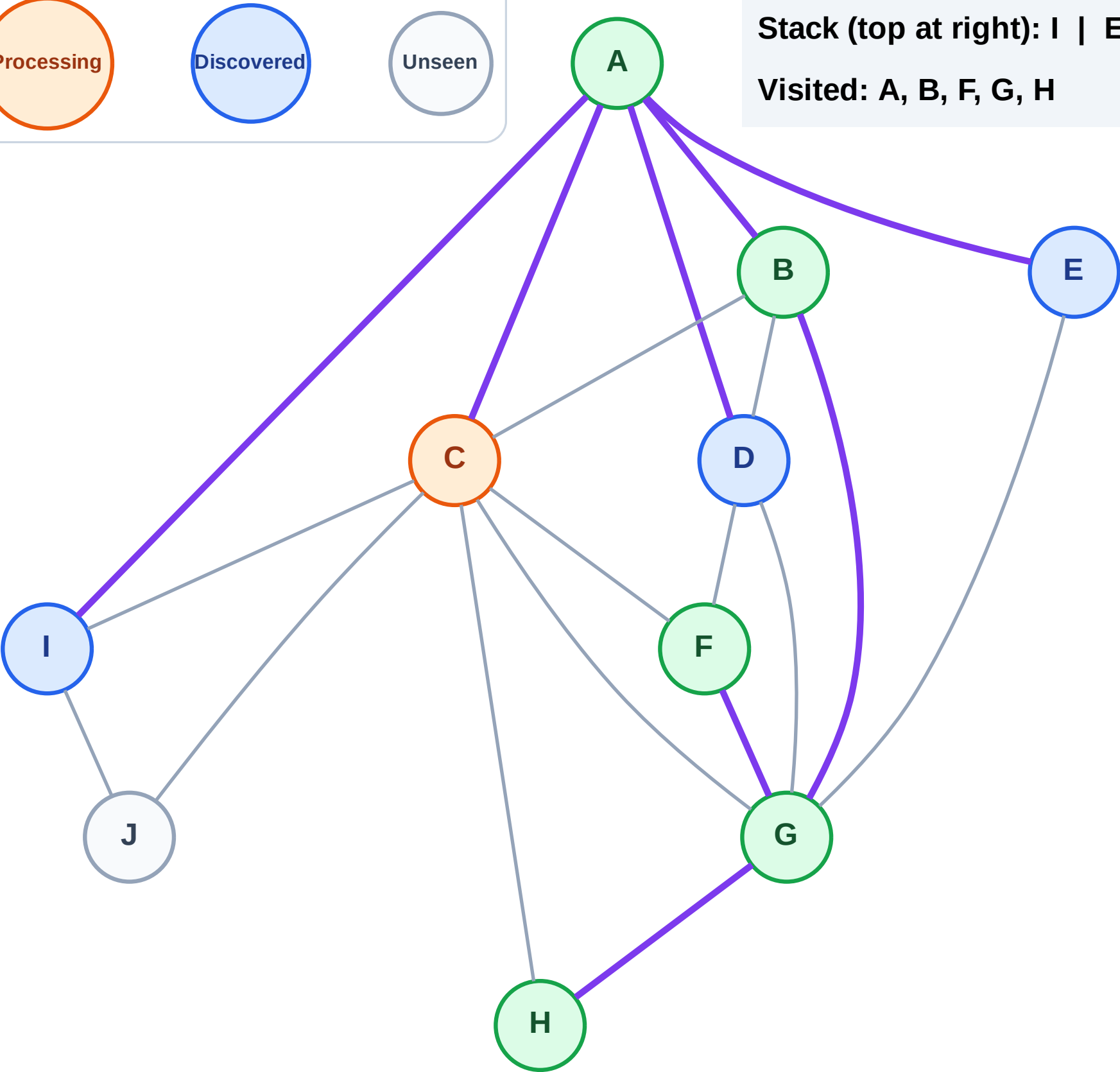
Complete

Processing

Discovered

Unseen

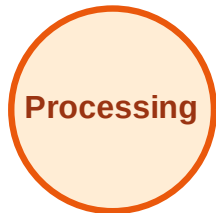
Stack (top at right): I | E | D
Visited: A, B, F, G, H



DFS Traversal

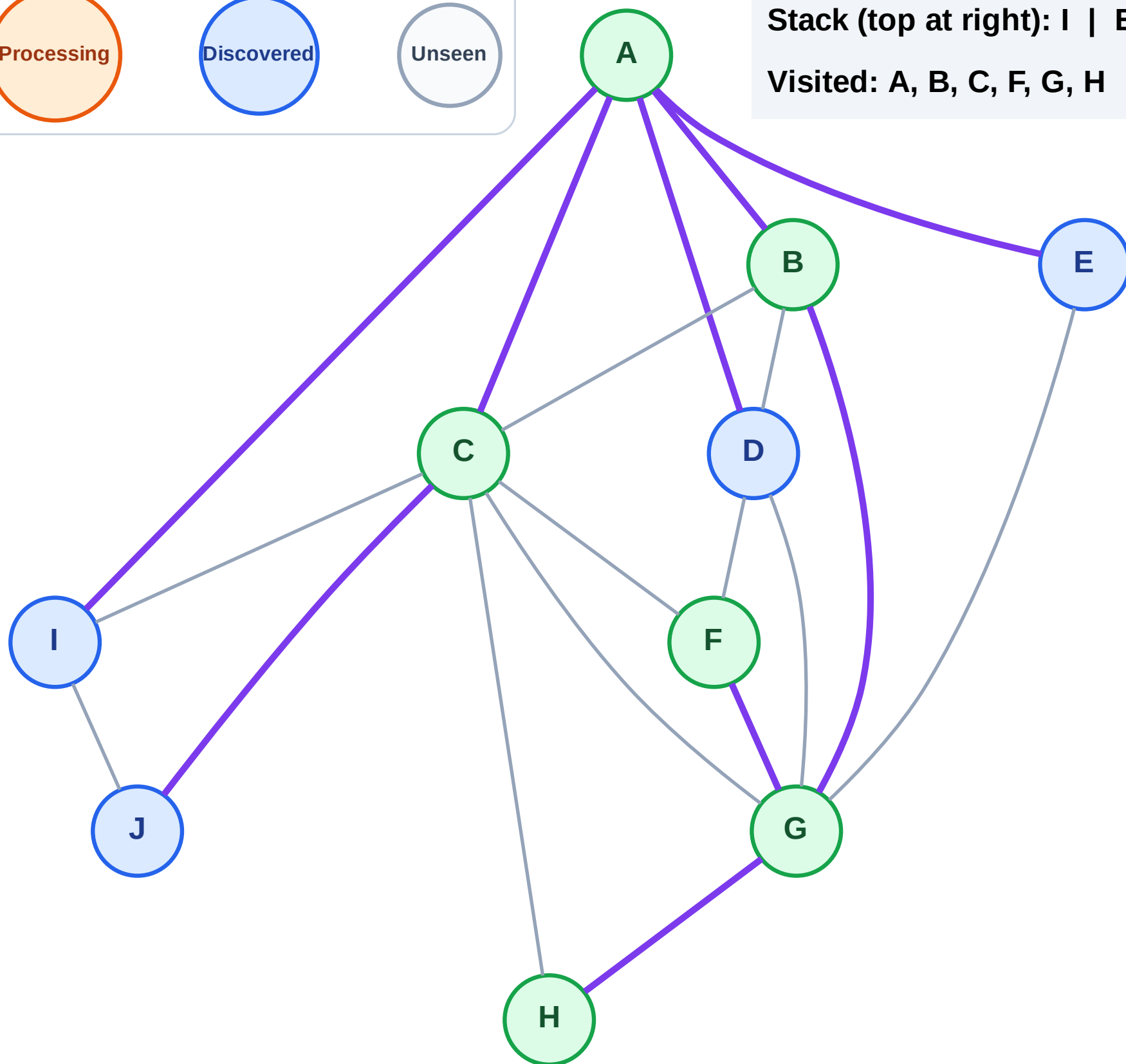
Step 13: Discover J from C

Legend



Stack (top at right): I | E | D | J

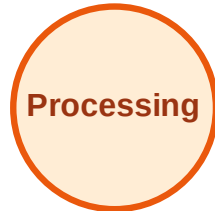
Visited: A, B, C, F, G, H



DFS Traversal

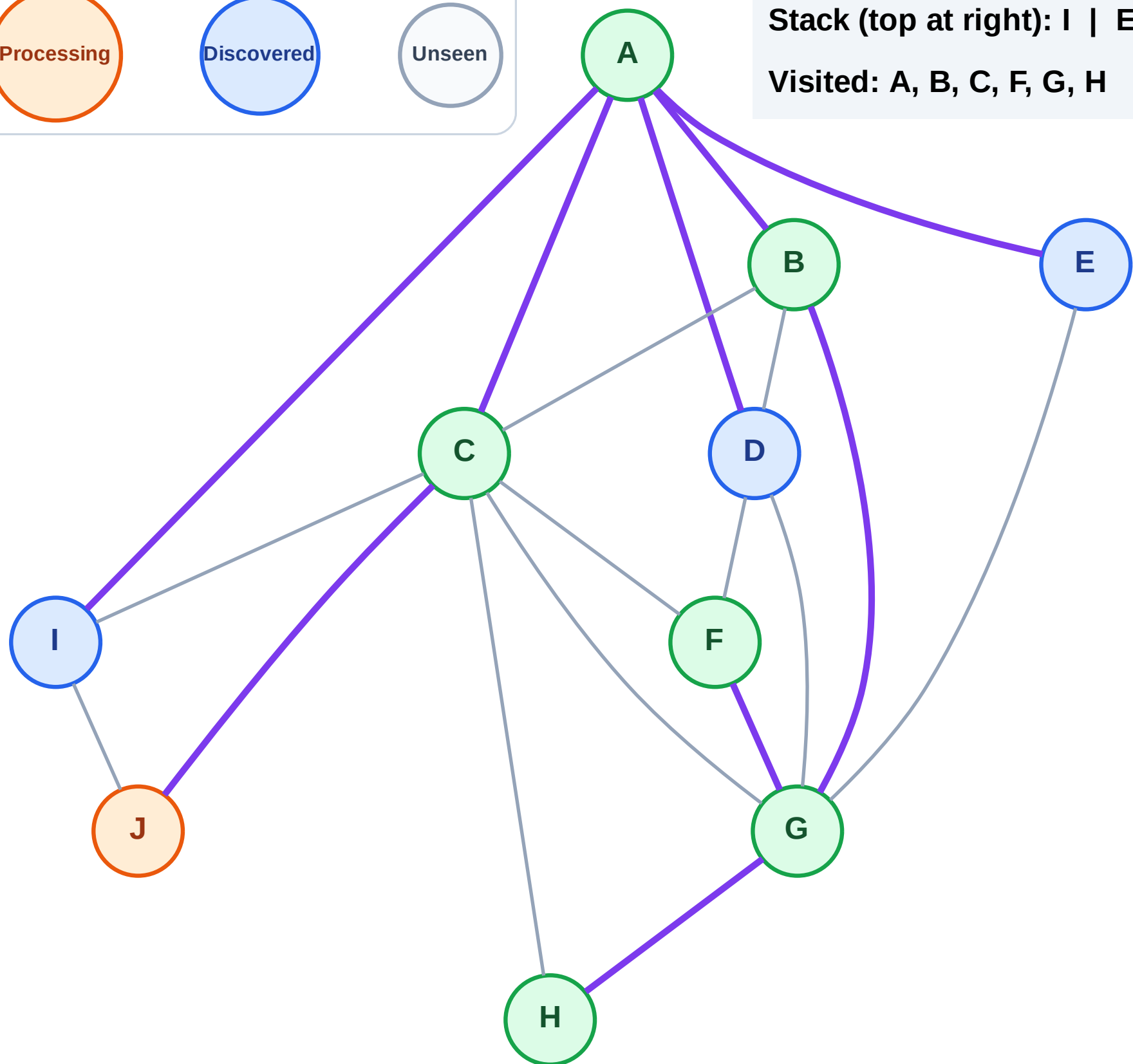
Step 14: Process node J

Legend



Stack (top at right): I | E | D

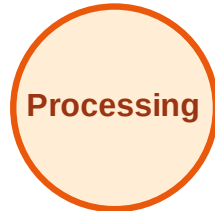
Visited: A, B, C, F, G, H



DFS Traversal

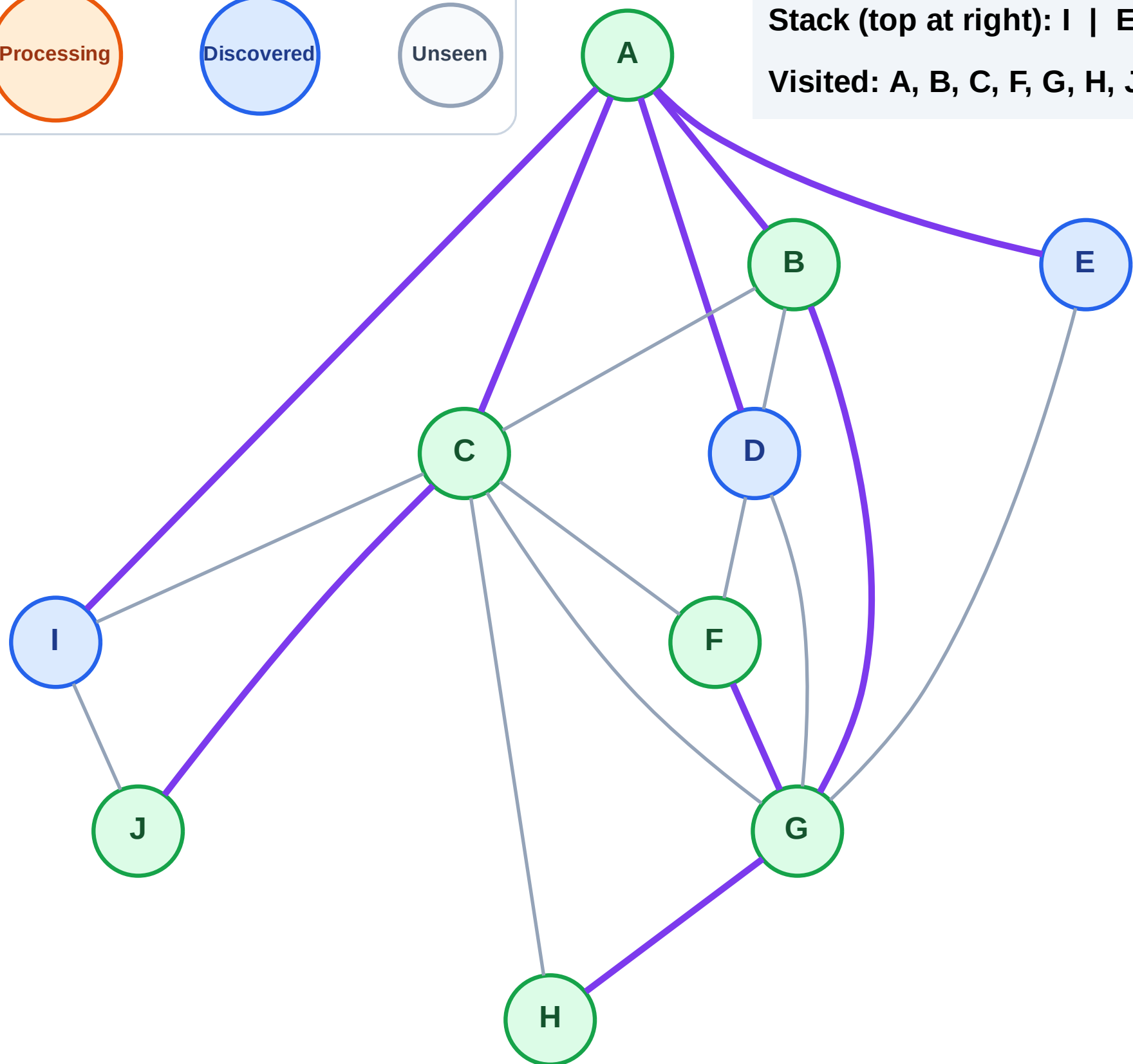
Step 15: No new neighbors from J

Legend



Stack (top at right): I | E | D

Visited: A, B, C, F, G, H, J



DFS Traversal

Step 16: Process node D

Legend

Complete

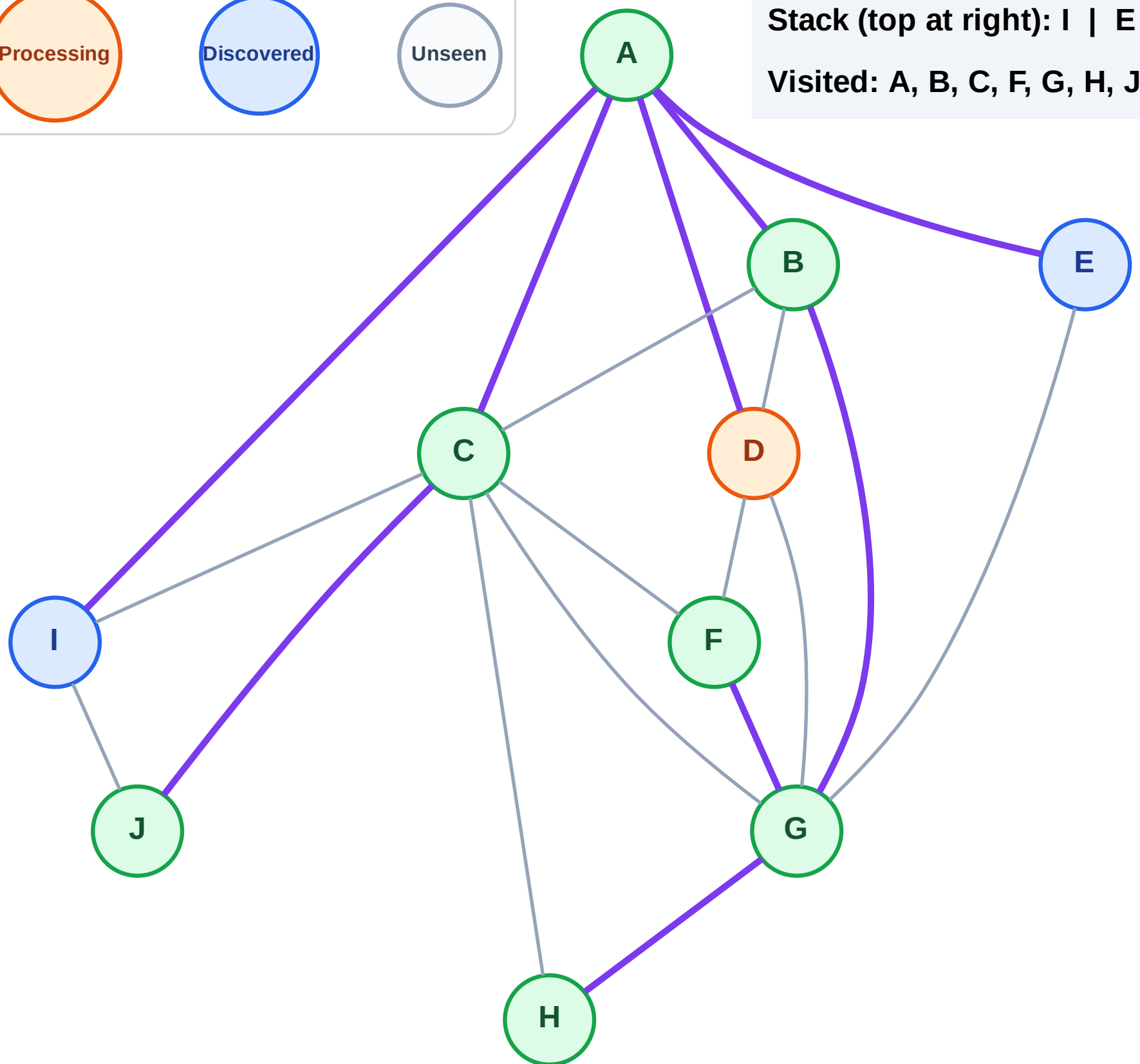
Processing

Discovered

Unseen

Stack (top at right): I | E

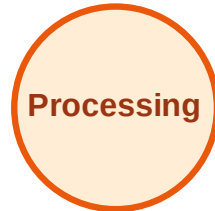
Visited: A, B, C, F, G, H, J



DFS Traversal

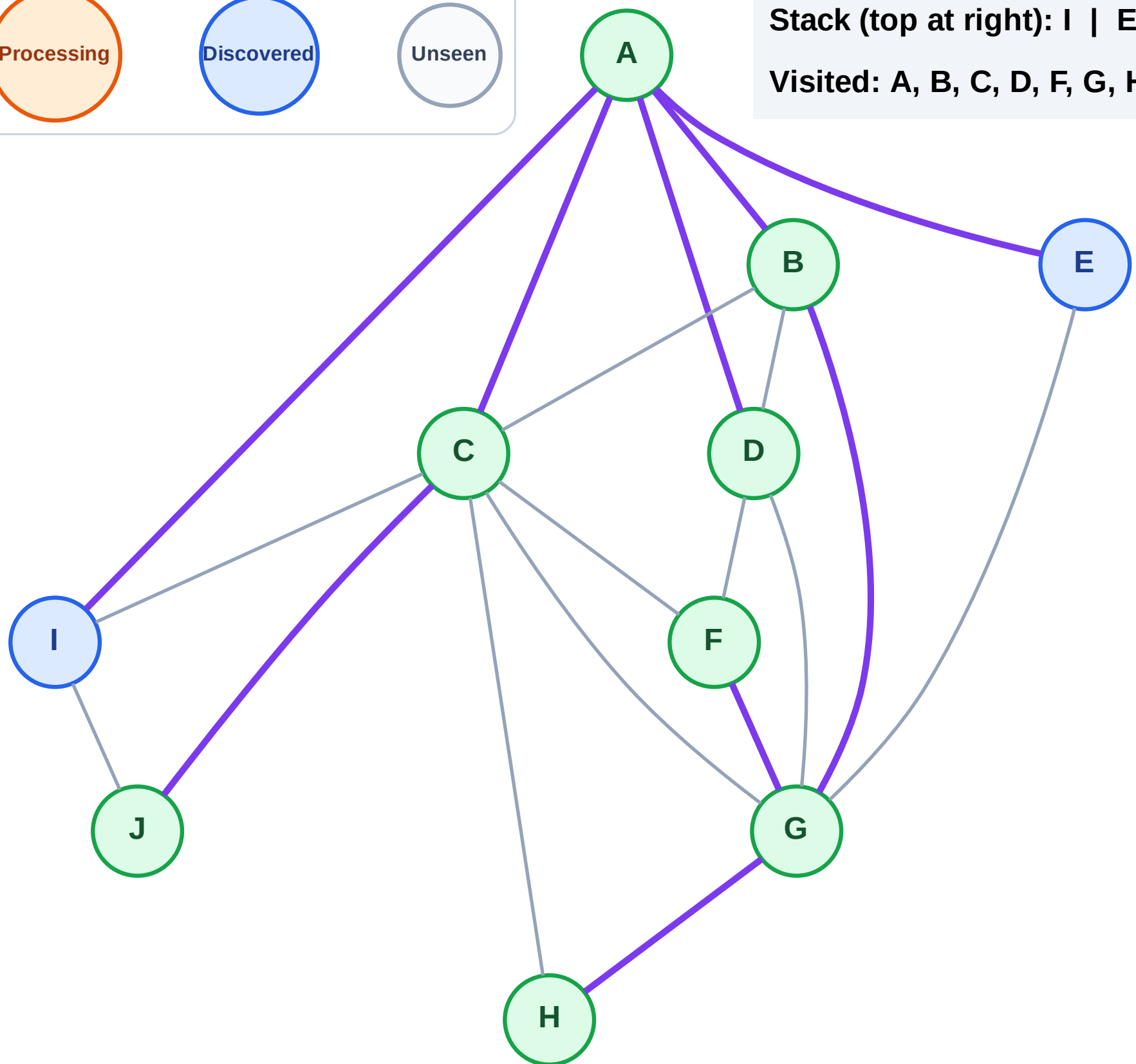
Step 17: No new neighbors from D

Legend



Stack (top at right): I | E

Visited: A, B, C, D, F, G, H, J



DFS Traversal

Step 18: Process node E

Legend

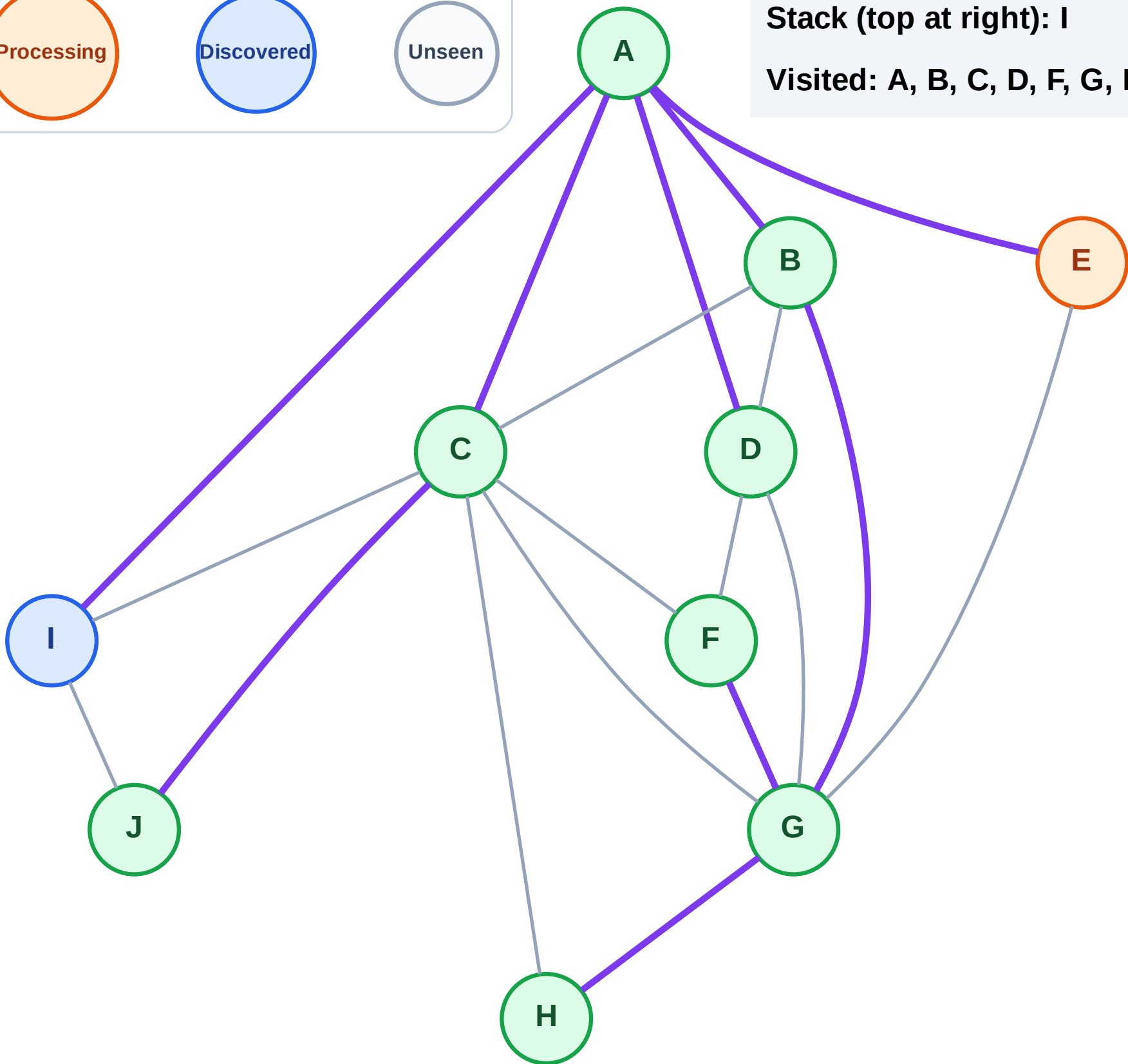
Complete

Processing

Discovered

Unseen

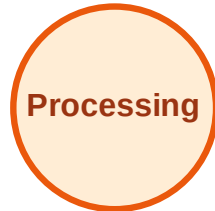
Stack (top at right): I
Visited: A, B, C, D, F, G, H, J



DFS Traversal

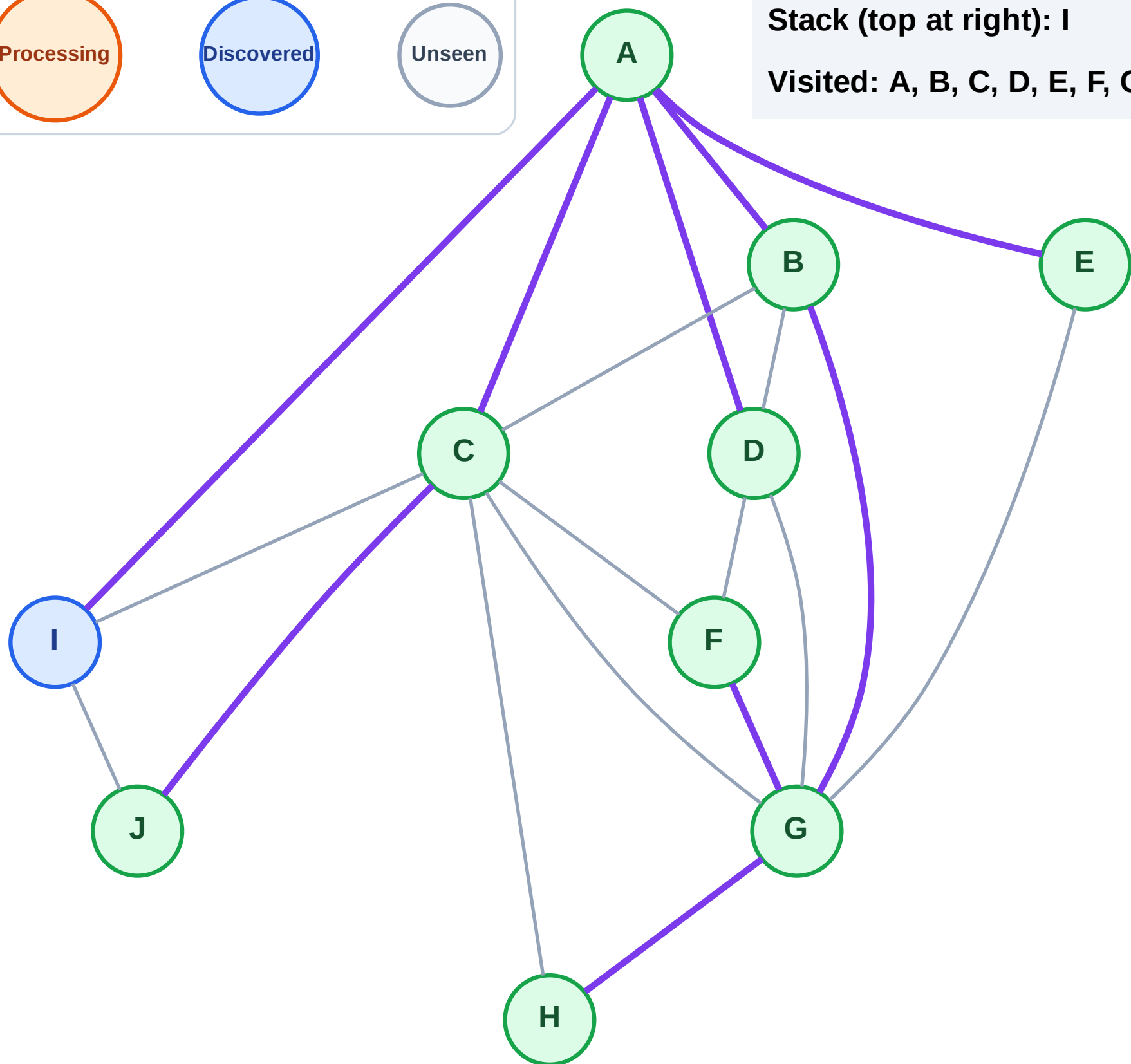
Step 19: No new neighbors from E

Legend



Stack (top at right): I

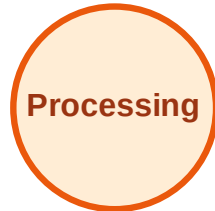
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

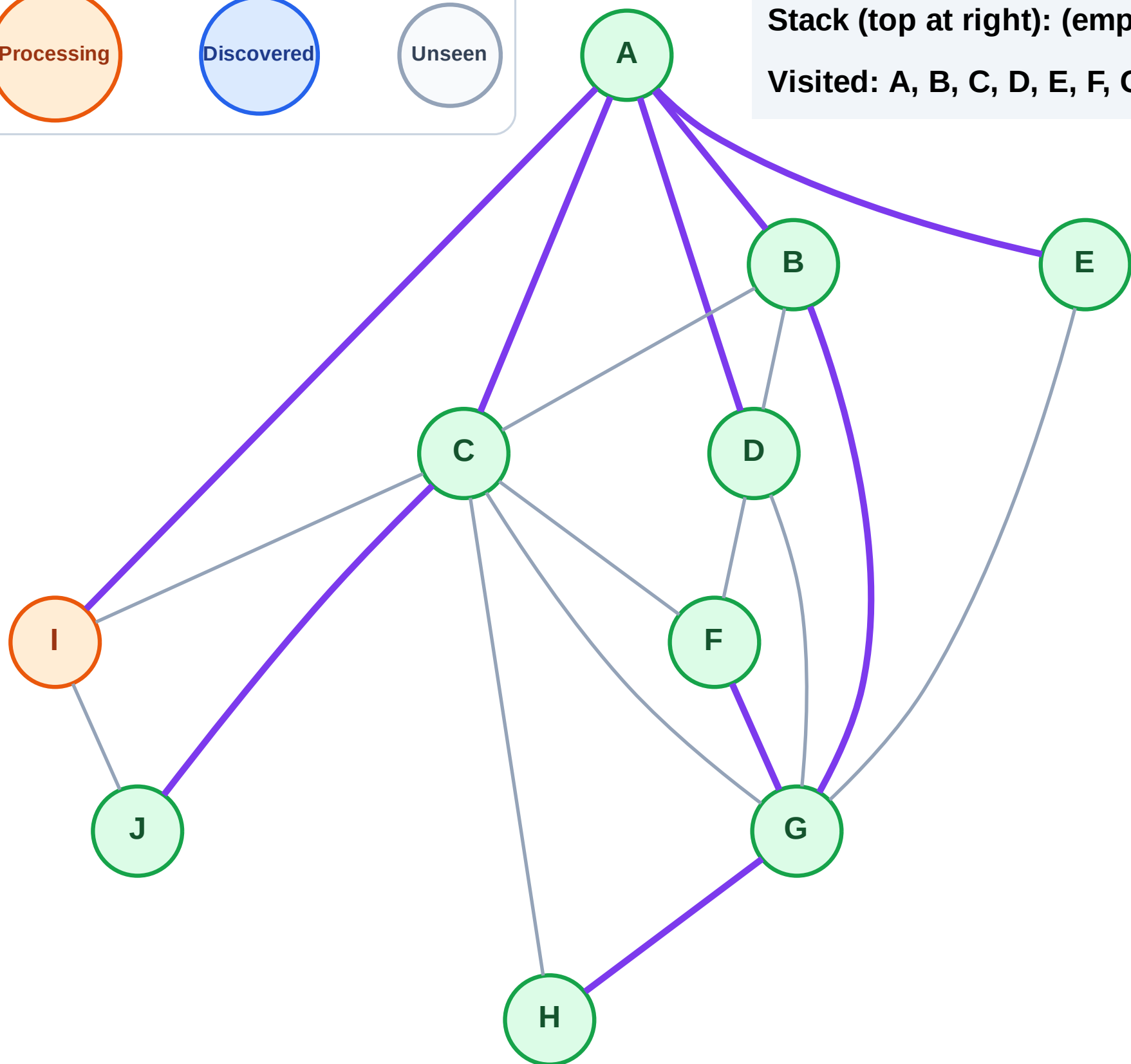
Step 20: Process node I

Legend



Stack (top at right): (empty)

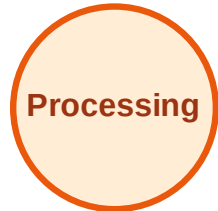
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

